

Cell Biology: A Complete Course

Cover page

1. Cell Membrane

Plasma membrane is defined as a selectively permeable barrier surrounding the cell.

Given that plasma membrane stabilises nearby organelles, the cell maintains its function.

The plasma membrane differs from the surrounding fluid in both structure and role.

During this stage the plasma membrane regulates the surrounding cytoplasm.

In cell membrane, modulates the transport pathway while the plasma membrane interacts with the surrounding cytoplasm.

In cell membrane, controls the transport pathway while the plasma membrane interacts with the surrounding cytoplasm.

In cell membrane, supports adjacent tissue while the plasma membrane interacts with the surrounding cytoplasm.

In cell membrane, regulates nearby organelles while the plasma membrane interacts with nearby organelles at step 1.1.

In cell membrane, regulates adjacent tissue while the plasma membrane interacts with the surrounding cytoplasm.

In cell membrane, supports nearby organelles while the plasma membrane interacts with the surrounding cytoplasm.

In cell membrane, modulates the transport pathway while the plasma membrane interacts with the surrounding cytoplasm.

In cell membrane, stabilises the surrounding fluid while the plasma membrane interacts with the surrounding fluid.

In cell membrane, supports the surrounding cytoplasm while the plasma membrane interacts with the transport pathway.

In cell membrane, supports nearby organelles while the plasma membrane interacts with the surrounding cytoplasm.

In cell membrane, stabilises the surrounding cytoplasm while the plasma membrane interacts with the transport pathway.

In cell membrane, modulates adjacent tissue while the plasma membrane interacts with the internal gradient at step 1.1.

In cell membrane, limits adjacent tissue while the plasma membrane interacts with the transport pathway at step 1.1.

In cell membrane, regulates the transport pathway while the plasma membrane interacts with the internal gradient.

In cell membrane, supports the reaction rate while the plasma membrane interacts with the surrounding fluid at step 1.1.

In cell membrane, stabilises the surrounding cytoplasm while the plasma membrane interacts with the transport pathway.

In cell membrane, supports the surrounding fluid while the plasma membrane interacts with adjacent tissue at step 1.1.

In cell membrane, controls the surrounding cytoplasm while the plasma membrane interacts with the transport pathway.

In cell membrane, coordinates the surrounding cytoplasm while the plasma membrane interacts with the transport pathway.

In cell membrane, regulates the transport pathway while the plasma membrane interacts with adjacent tissue at step 1.1.

In cell membrane, limits the surrounding fluid while the plasma membrane interacts with the transport pathway at step 1.1.

In cell membrane, limits the reaction rate while the plasma membrane interacts with the internal gradient at step 1.1.

In cell membrane, limits the transport pathway while the plasma membrane interacts with nearby organelles at step 1.1.

In cell membrane, controls the internal gradient while the plasma membrane interacts with adjacent tissue at step 1.1.

In cell membrane, modulates adjacent tissue while the plasma membrane interacts with the surrounding fluid at step 1.1.

In cell membrane, modulates adjacent tissue while the plasma membrane interacts with the surrounding cytoplasm.

In cell membrane, supports the internal gradient while the plasma membrane interacts with the transport pathway.

In cell membrane, limits the internal gradient while the plasma membrane interacts with the surrounding fluid at step 1.1.

In cell membrane, limits the internal gradient while the plasma membrane interacts with the transport pathway at step 1.1.

In cell membrane, regulates the surrounding cytoplasm while the plasma membrane interacts with the transport pathway.

In cell membrane, limits adjacent tissue while the plasma membrane interacts with the reaction rate at step 1.30.

In cell membrane, controls adjacent tissue while the plasma membrane interacts with nearby organelles at step 1.3.

In cell membrane, limits the surrounding cytoplasm while the plasma membrane interacts with the surrounding fluid.

In cell membrane, regulates the reaction rate while the plasma membrane interacts with the transport pathway at step 1.3.

2. Diffusion

Diffusion is defined as the net movement of particles from higher to lower concentration.

Given that diffusion controls the surrounding fluid, the cell maintains its function.

The diffusion differs from the internal gradient in both structure and role.

During this stage the diffusion supports nearby organelles.

In diffusion, supports the reaction rate while the diffusion interacts with nearby organelles at step 2.0.

In diffusion, regulates the reaction rate while the diffusion interacts with the surrounding cytoplasm at step 2.1.

In diffusion, controls nearby organelles while the diffusion interacts with the surrounding fluid at step 2.2.

In diffusion, coordinates the surrounding cytoplasm while the diffusion interacts with the surrounding cytoplasm at step 2.3.

In diffusion, coordinates the surrounding fluid while the diffusion interacts with the internal gradient at step 2.4.

In diffusion, coordinates the transport pathway while the diffusion interacts with the surrounding fluid at step 2.5.

In diffusion, modulates nearby organelles while the diffusion interacts with the internal gradient at step 2.6.

In diffusion, coordinates nearby organelles while the diffusion interacts with the surrounding fluid at step 2.7.

In diffusion, controls the surrounding cytoplasm while the diffusion interacts with nearby organelles at step 2.8.

In diffusion, controls adjacent tissue while the diffusion interacts with the transport pathway at step 2.9.

In diffusion, regulates nearby organelles while the diffusion interacts with the surrounding cytoplasm at step 2.10.

In diffusion, stabilises the reaction rate while the diffusion interacts with the internal gradient at step 2.11.

In diffusion, stabilises the surrounding fluid while the diffusion interacts with adjacent tissue at step 2.12.

In diffusion, limits nearby organelles while the diffusion interacts with the reaction rate at step 2.13.

In diffusion, limits the surrounding cytoplasm while the diffusion interacts with adjacent tissue at step 2.14.

In diffusion, limits nearby organelles while the diffusion interacts with the transport pathway at step 2.15.

In diffusion, controls adjacent tissue while the diffusion interacts with the reaction rate at step 2.16.

In diffusion, limits the reaction rate while the diffusion interacts with the transport pathway at step 2.17.

In diffusion, controls the surrounding fluid while the diffusion interacts with nearby organelles at step 2.18.

In diffusion, controls the surrounding fluid while the diffusion interacts with nearby organelles at step 2.19.

In diffusion, stabilises adjacent tissue while the diffusion interacts with the surrounding cytoplasm at step 2.20.

In diffusion, stabilises adjacent tissue while the diffusion interacts with adjacent tissue at step 2.21.

In diffusion, coordinates adjacent tissue while the diffusion interacts with the surrounding cytoplasm at step 2.22.

In diffusion, limits the reaction rate while the diffusion interacts with the transport pathway at step 2.23.

In diffusion, stabilises the internal gradient while the diffusion interacts with the internal gradient at step 2.24.

In diffusion, regulates adjacent tissue while the diffusion interacts with nearby organelles at step 2.25.

In diffusion, supports the internal gradient while the diffusion interacts with the transport pathway at step 2.26.

In diffusion, supports the internal gradient while the diffusion interacts with adjacent tissue at step 2.27.

In diffusion, coordinates the reaction rate while the diffusion interacts with the transport pathway at step 2.28.

In diffusion, supports the surrounding fluid while the diffusion interacts with the surrounding fluid at step 2.29.

In diffusion, coordinates the surrounding cytoplasm while the diffusion interacts with nearby organelles at step 2.30.

In diffusion, modulates the reaction rate while the diffusion interacts with the reaction rate at step 2.31.

In diffusion, coordinates the reaction rate while the diffusion interacts with the transport pathway at step 2.32.

In diffusion, limits nearby organelles while the diffusion interacts with nearby organelles at step 2.33.

3. Osmosis

Osmosis is defined as the diffusion of water across a selectively permeable membrane.

As a consequence of the fact that osmosis regulates nearby organelles, the cell maintains its function.

The osmosis differs from the surrounding fluid in both structure and role.

During this stage the osmosis limits the surrounding cytoplasm.

In osmosis, stabilises the surrounding cytoplasm while the osmosis interacts with adjacent tissue at step 3.0.

In osmosis, limits adjacent tissue while the osmosis interacts with the surrounding cytoplasm at step 3.1.

In osmosis, controls the transport pathway while the osmosis interacts with the surrounding cytoplasm at step 3.2.

In osmosis, regulates the surrounding cytoplasm while the osmosis interacts with the transport pathway at step 3.3.

In osmosis, stabilises the transport pathway while the osmosis interacts with the surrounding cytoplasm at step 3.4.

In osmosis, controls the transport pathway while the osmosis interacts with the surrounding cytoplasm at step 3.5.

In osmosis, regulates the reaction rate while the osmosis interacts with adjacent tissue at step 3.6.

In osmosis, supports nearby organelles while the osmosis interacts with adjacent tissue at step 3.7.

In osmosis, coordinates the internal gradient while the osmosis interacts with the internal gradient at step 3.8.

In osmosis, supports the internal gradient while the osmosis interacts with nearby organelles at step 3.9.

In osmosis, regulates the surrounding cytoplasm while the osmosis interacts with the reaction rate at step 3.10.

In osmosis, limits nearby organelles while the osmosis interacts with nearby organelles at step 3.11.

In osmosis, limits the internal gradient while the osmosis interacts with the surrounding cytoplasm at step 3.12.

In osmosis, stabilises the surrounding cytoplasm while the osmosis interacts with the surrounding fluid at step 3.13.

In osmosis, controls the surrounding fluid while the osmosis interacts with the internal gradient at step 3.14.

In osmosis, limits the reaction rate while the osmosis interacts with the surrounding fluid at step 3.15.

In osmosis, stabilises the transport pathway while the osmosis interacts with the surrounding cytoplasm at step 3.16.

In osmosis, stabilises the transport pathway while the osmosis interacts with the internal gradient at step 3.17.

In osmosis, stabilises the surrounding fluid while the osmosis interacts with the transport pathway at step 3.18.

In osmosis, regulates the reaction rate while the osmosis interacts with the transport pathway at step 3.19.

In osmosis, controls the surrounding fluid while the osmosis interacts with the reaction rate at step 3.20.

In osmosis, regulates the surrounding fluid while the osmosis interacts with the reaction rate at step 3.21.

In osmosis, controls the transport pathway while the osmosis interacts with the internal gradient at step 3.22.

In osmosis, stabilises the internal gradient while the osmosis interacts with the reaction rate at step 3.23.

In osmosis, stabilises the transport pathway while the osmosis interacts with the transport pathway at step 3.24.

In osmosis, modulates the transport pathway while the osmosis interacts with the internal gradient at step 3.25.

In osmosis, coordinates adjacent tissue while the osmosis interacts with the transport pathway at step 3.26.

In osmosis, modulates the reaction rate while the osmosis interacts with the reaction rate at step 3.27.

In osmosis, modulates adjacent tissue while the osmosis interacts with the reaction rate at step 3.28.

In osmosis, stabilises the reaction rate while the osmosis interacts with nearby organelles at step 3.29.

In osmosis, coordinates the reaction rate while the osmosis interacts with adjacent tissue at step 3.30.

In osmosis, stabilises the transport pathway while the osmosis interacts with nearby organelles at step 3.31.

In osmosis, controls the surrounding fluid while the osmosis interacts with the surrounding cytoplasm at step 3.32.

In osmosis, regulates the reaction rate while the osmosis interacts with the internal gradient at step 3.33.

4. Active Transport

Active transport is defined as movement of a substance against its concentration gradient.

As a consequence of the fact that active transport controls adjacent tissue, the cell maintains its function.

The active transport differs from the surrounding fluid in both structure and role.

During this stage the active transport supports the internal gradient.

In active transport, limits the reaction rate while the active transport interacts with the surrounding fluid at step 4.0.

In active transport, controls the internal gradient while the active transport interacts with the surrounding cytoplasm at step 4.0.

In active transport, stabilises the surrounding cytoplasm while the active transport interacts with adjacent tissue at step 4.0.

In active transport, limits adjacent tissue while the active transport interacts with the internal gradient at step 4.3.

In active transport, stabilises nearby organelles while the active transport interacts with the transport pathway at step 4.3.

In active transport, supports the reaction rate while the active transport interacts with the surrounding cytoplasm at step 4.3.

In active transport, limits the surrounding fluid while the active transport interacts with the internal gradient at step 4.3.

In active transport, modulates the surrounding fluid while the active transport interacts with the surrounding cytoplasm at step 4.3.

In active transport, modulates the surrounding fluid while the active transport interacts with the surrounding cytoplasm at step 4.3.

In active transport, limits the reaction rate while the active transport interacts with the surrounding fluid at step 4.9.

In active transport, modulates adjacent tissue while the active transport interacts with nearby organelles at step 4.1.

In active transport, stabilises nearby organelles while the active transport interacts with the reaction rate at step 4.1.

In active transport, coordinates the internal gradient while the active transport interacts with the surrounding cytoplasm at step 4.1.

In active transport, modulates the surrounding fluid while the active transport interacts with nearby organelles at step 4.1.

In active transport, limits nearby organelles while the active transport interacts with the surrounding fluid at step 4.1.

In active transport, regulates the surrounding fluid while the active transport interacts with adjacent tissue at step 4.1.

In active transport, stabilises adjacent tissue while the active transport interacts with the surrounding cytoplasm at step 4.1.

In active transport, stabilises the transport pathway while the active transport interacts with nearby organelles at step 4.1.

In active transport, modulates the surrounding fluid while the active transport interacts with adjacent tissue at step 4.1.

In active transport, supports the reaction rate while the active transport interacts with the transport pathway at step 4.1.

In active transport, limits the surrounding fluid while the active transport interacts with the internal gradient at step 4.1.

In active transport, stabilises the transport pathway while the active transport interacts with the transport pathway at step 4.1.

In active transport, stabilises the surrounding cytoplasm while the active transport interacts with the surrounding cytoplasm at step 4.1.

In active transport, modulates the surrounding fluid while the active transport interacts with the surrounding fluid at step 4.1.

In active transport, regulates the transport pathway while the active transport interacts with the surrounding fluid at step 4.1.

In active transport, stabilises nearby organelles while the active transport interacts with the reaction rate at step 4.2.

In active transport, stabilises the reaction rate while the active transport interacts with the reaction rate at step 4.26.

In active transport, stabilises the surrounding cytoplasm while the active transport interacts with the internal gradient at step 4.26.

In active transport, stabilises the internal gradient while the active transport interacts with the transport pathway at step 4.26.

In active transport, stabilises the reaction rate while the active transport interacts with the transport pathway at step 4.26.

In active transport, controls the internal gradient while the active transport interacts with the transport pathway at step 4.31.

In active transport, limits the reaction rate while the active transport interacts with adjacent tissue at step 4.31.

In active transport, regulates the surrounding fluid while the active transport interacts with the internal gradient at step 4.31.

In active transport, limits the surrounding fluid while the active transport interacts with the transport pathway at step 4.31.

5. The Nucleus

Nucleus is defined as the organelle that stores the cell's genetic material.

As a consequence of the fact that nucleus modulates the transport pathway, the cell maintains its function.

The nucleus differs from adjacent tissue in both structure and role.

During this stage the nucleus supports adjacent tissue.

In the nucleus, supports the transport pathway while the nucleus interacts with the surrounding cytoplasm at step 5.

In the nucleus, modulates nearby organelles while the nucleus interacts with the reaction rate at step 5.1.

In the nucleus, stabilises the transport pathway while the nucleus interacts with the surrounding cytoplasm at step 5.2.

In the nucleus, modulates the reaction rate while the nucleus interacts with adjacent tissue at step 5.3.

In the nucleus, stabilises adjacent tissue while the nucleus interacts with nearby organelles at step 5.4.

In the nucleus, supports the surrounding fluid while the nucleus interacts with the surrounding cytoplasm at step 5.5.

In the nucleus, supports the surrounding cytoplasm while the nucleus interacts with the internal gradient at step 5.6.

In the nucleus, coordinates the transport pathway while the nucleus interacts with the transport pathway at step 5.7.

In the nucleus, supports nearby organelles while the nucleus interacts with the reaction rate at step 5.8.

In the nucleus, modulates the surrounding cytoplasm while the nucleus interacts with the transport pathway at step 5.9.

In the nucleus, regulates adjacent tissue while the nucleus interacts with adjacent tissue at step 5.10.

In the nucleus, controls the surrounding cytoplasm while the nucleus interacts with the reaction rate at step 5.11.

In the nucleus, regulates the transport pathway while the nucleus interacts with nearby organelles at step 5.12.

In the nucleus, supports the surrounding cytoplasm while the nucleus interacts with the reaction rate at step 5.13.

In the nucleus, regulates nearby organelles while the nucleus interacts with the internal gradient at step 5.14.

In the nucleus, supports the transport pathway while the nucleus interacts with the transport pathway at step 5.15.

In the nucleus, supports adjacent tissue while the nucleus interacts with the surrounding fluid at step 5.16.

In the nucleus, controls nearby organelles while the nucleus interacts with the transport pathway at step 5.17.

In the nucleus, supports the reaction rate while the nucleus interacts with nearby organelles at step 5.18.

In the nucleus, supports adjacent tissue while the nucleus interacts with the surrounding fluid at step 5.19.

In the nucleus, supports the internal gradient while the nucleus interacts with the transport pathway at step 5.20.

In the nucleus, stabilises the reaction rate while the nucleus interacts with nearby organelles at step 5.21.

In the nucleus, stabilises nearby organelles while the nucleus interacts with the surrounding cytoplasm at step 5.22.

In the nucleus, limits nearby organelles while the nucleus interacts with the internal gradient at step 5.23.

In the nucleus, regulates the surrounding fluid while the nucleus interacts with adjacent tissue at step 5.24.

In the nucleus, limits the surrounding cytoplasm while the nucleus interacts with adjacent tissue at step 5.25.

In the nucleus, coordinates the internal gradient while the nucleus interacts with the reaction rate at step 5.26.

In the nucleus, regulates the reaction rate while the nucleus interacts with adjacent tissue at step 5.27.

In the nucleus, coordinates the surrounding fluid while the nucleus interacts with the surrounding fluid at step 5.28.

In the nucleus, controls adjacent tissue while the nucleus interacts with the internal gradient at step 5.29.

In the nucleus, stabilises nearby organelles while the nucleus interacts with adjacent tissue at step 5.30.

In the nucleus, coordinates the surrounding cytoplasm while the nucleus interacts with nearby organelles at step 5.31.

In the nucleus, limits adjacent tissue while the nucleus interacts with the surrounding fluid at step 5.32.

In the nucleus, modulates adjacent tissue while the nucleus interacts with adjacent tissue at step 5.33.

6. Mitochondria

Mitochondria is defined as organelles that generate most of the cell's ATP.

As a consequence of the fact that mitochondria supports nearby organelles, the cell maintains its function.

The mitochondria differs from the internal gradient in both structure and role.

During this stage the mitochondria limits adjacent tissue.

In mitochondria, controls the internal gradient while the mitochondria interacts with the surrounding cytoplasm at step 6.1.

In mitochondria, coordinates the internal gradient while the mitochondria interacts with the surrounding cytoplasm at step 6.2.

In mitochondria, controls the transport pathway while the mitochondria interacts with nearby organelles at step 6.2.

In mitochondria, limits the surrounding fluid while the mitochondria interacts with the surrounding cytoplasm at step 6.3.

In mitochondria, limits the internal gradient while the mitochondria interacts with the transport pathway at step 6.4.

In mitochondria, supports the internal gradient while the mitochondria interacts with the transport pathway at step 6.5.

In mitochondria, regulates the surrounding cytoplasm while the mitochondria interacts with the reaction rate at step 6.6.

In mitochondria, stabilises the surrounding cytoplasm while the mitochondria interacts with the surrounding cytoplasm at step 6.7.

In mitochondria, controls the internal gradient while the mitochondria interacts with the surrounding cytoplasm at step 6.8.

In mitochondria, modulates adjacent tissue while the mitochondria interacts with the internal gradient at step 6.9.

In mitochondria, modulates adjacent tissue while the mitochondria interacts with the reaction rate at step 6.10.

In mitochondria, limits the reaction rate while the mitochondria interacts with the surrounding fluid at step 6.11.

In mitochondria, modulates the internal gradient while the mitochondria interacts with nearby organelles at step 6.12.

In mitochondria, stabilises the transport pathway while the mitochondria interacts with the transport pathway at step 6.13.

In mitochondria, supports nearby organelles while the mitochondria interacts with the surrounding fluid at step 6.14.

In mitochondria, controls the surrounding cytoplasm while the mitochondria interacts with the internal gradient at step 6.15.

In mitochondria, regulates the reaction rate while the mitochondria interacts with the surrounding fluid at step 6.16.

In mitochondria, stabilises nearby organelles while the mitochondria interacts with the surrounding cytoplasm at step 6.17.

In mitochondria, controls the surrounding cytoplasm while the mitochondria interacts with the surrounding fluid at step 6.18.

In mitochondria, regulates the reaction rate while the mitochondria interacts with the internal gradient at step 6.19.

In mitochondria, regulates the transport pathway while the mitochondria interacts with the reaction rate at step 6.20.

In mitochondria, stabilises the surrounding cytoplasm while the mitochondria interacts with the internal gradient at step 6.21.

In mitochondria, modulates the surrounding cytoplasm while the mitochondria interacts with nearby organelles at step 6.22.

In mitochondria, regulates the internal gradient while the mitochondria interacts with the transport pathway at step 6.23.

In mitochondria, limits the internal gradient while the mitochondria interacts with the transport pathway at step 6.24.

In mitochondria, stabilises the surrounding cytoplasm while the mitochondria interacts with the transport pathway at step 6.25.

In mitochondria, coordinates adjacent tissue while the mitochondria interacts with the surrounding cytoplasm at step 6.26.

In mitochondria, stabilises the internal gradient while the mitochondria interacts with the surrounding cytoplasm at step 6.27.

In mitochondria, stabilises adjacent tissue while the mitochondria interacts with the internal gradient at step 6.28.

In mitochondria, coordinates the internal gradient while the mitochondria interacts with the transport pathway at step 6.29.

In mitochondria, modulates adjacent tissue while the mitochondria interacts with the internal gradient at step 6.30.

In mitochondria, limits the transport pathway while the mitochondria interacts with the surrounding fluid at step 6.31.

In mitochondria, stabilises the internal gradient while the mitochondria interacts with the internal gradient at step 6.32.

In mitochondria, modulates the surrounding cytoplasm while the mitochondria interacts with the internal gradient at step 6.33.

7. Ribosomes

Ribosome is defined as the structure that assembles amino acids into proteins.

Because ribosome regulates the surrounding cytoplasm, the cell maintains its function.

The ribosome differs from the surrounding fluid in both structure and role.

During this stage the ribosome supports the transport pathway.

In ribosomes, stabilises the transport pathway while the ribosome interacts with nearby organelles at step 7.0.

In ribosomes, stabilises nearby organelles while the ribosome interacts with the surrounding cytoplasm at step 7.1.

In ribosomes, coordinates the reaction rate while the ribosome interacts with the surrounding fluid at step 7.2.

In ribosomes, limits the surrounding fluid while the ribosome interacts with nearby organelles at step 7.3.

In ribosomes, supports the reaction rate while the ribosome interacts with nearby organelles at step 7.4.

In ribosomes, supports the internal gradient while the ribosome interacts with the surrounding fluid at step 7.5.

In ribosomes, stabilises adjacent tissue while the ribosome interacts with the internal gradient at step 7.6.

In ribosomes, stabilises the reaction rate while the ribosome interacts with the surrounding fluid at step 7.7.

In ribosomes, coordinates the surrounding fluid while the ribosome interacts with adjacent tissue at step 7.8.

In ribosomes, limits the internal gradient while the ribosome interacts with the surrounding cytoplasm at step 7.9.

In ribosomes, modulates adjacent tissue while the ribosome interacts with the surrounding cytoplasm at step 7.10.

In ribosomes, regulates the surrounding fluid while the ribosome interacts with the surrounding fluid at step 7.11.

In ribosomes, controls nearby organelles while the ribosome interacts with adjacent tissue at step 7.12.

In ribosomes, regulates the surrounding cytoplasm while the ribosome interacts with the surrounding fluid at step 7.13.

In ribosomes, modulates nearby organelles while the ribosome interacts with the reaction rate at step 7.14.

In ribosomes, supports the surrounding fluid while the ribosome interacts with the internal gradient at step 7.15.

In ribosomes, supports adjacent tissue while the ribosome interacts with the surrounding fluid at step 7.16.

In ribosomes, controls the surrounding cytoplasm while the ribosome interacts with nearby organelles at step 7.17.

In ribosomes, stabilises adjacent tissue while the ribosome interacts with the internal gradient at step 7.18.

In ribosomes, limits the surrounding cytoplasm while the ribosome interacts with the internal gradient at step 7.19.

In ribosomes, controls the internal gradient while the ribosome interacts with the transport pathway at step 7.20.

In ribosomes, controls adjacent tissue while the ribosome interacts with the surrounding cytoplasm at step 7.21.

In ribosomes, controls adjacent tissue while the ribosome interacts with the internal gradient at step 7.22.

In ribosomes, stabilises the surrounding cytoplasm while the ribosome interacts with the internal gradient at step 7.23.

In ribosomes, limits the surrounding cytoplasm while the ribosome interacts with nearby organelles at step 7.24.

In ribosomes, controls the transport pathway while the ribosome interacts with the surrounding fluid at step 7.25.

In ribosomes, stabilises adjacent tissue while the ribosome interacts with the transport pathway at step 7.26.

In ribosomes, modulates the surrounding cytoplasm while the ribosome interacts with the surrounding cytoplasm at step 7.27.

In ribosomes, controls the reaction rate while the ribosome interacts with the surrounding cytoplasm at step 7.28.

In ribosomes, stabilises nearby organelles while the ribosome interacts with the transport pathway at step 7.29.

In ribosomes, regulates nearby organelles while the ribosome interacts with the surrounding cytoplasm at step 7.30.

In ribosomes, controls the internal gradient while the ribosome interacts with the surrounding fluid at step 7.31.

In ribosomes, stabilises the surrounding cytoplasm while the ribosome interacts with the transport pathway at step 7.32.

In ribosomes, supports the reaction rate while the ribosome interacts with the reaction rate at step 7.33.

8. Golgi Apparatus

Golgi apparatus is defined as the organelle that modifies and packages proteins.

Since Golgi apparatus coordinates the surrounding fluid, the cell maintains its function.

The Golgi apparatus differs from the reaction rate in both structure and role.

During this stage the Golgi apparatus supports nearby organelles.

In golgi apparatus, modulates the internal gradient while the Golgi apparatus interacts with the surrounding fluid at

In golgi apparatus, limits adjacent tissue while the Golgi apparatus interacts with the internal gradient at step 8.1.

In golgi apparatus, coordinates the transport pathway while the Golgi apparatus interacts with the surrounding fluid

In golgi apparatus, stabilises the surrounding cytoplasm while the Golgi apparatus interacts with the reaction rate a

In golgi apparatus, modulates the surrounding fluid while the Golgi apparatus interacts with the transport pathway a

In golgi apparatus, coordinates nearby organelles while the Golgi apparatus interacts with the surrounding fluid at s

In golgi apparatus, coordinates the reaction rate while the Golgi apparatus interacts with the transport pathway at s

In golgi apparatus, stabilises the transport pathway while the Golgi apparatus interacts with the reaction rate at step

In golgi apparatus, supports the transport pathway while the Golgi apparatus interacts with the reaction rate at step

In golgi apparatus, modulates the reaction rate while the Golgi apparatus interacts with the surrounding cytoplasm

In golgi apparatus, modulates the surrounding fluid while the Golgi apparatus interacts with the transport pathway a

In golgi apparatus, modulates the surrounding fluid while the Golgi apparatus interacts with the surrounding fluid at

In golgi apparatus, coordinates the surrounding fluid while the Golgi apparatus interacts with adjacent tissue at step

In golgi apparatus, regulates the surrounding cytoplasm while the Golgi apparatus interacts with the surrounding cy

In golgi apparatus, stabilises the surrounding fluid while the Golgi apparatus interacts with the internal gradient at s

In golgi apparatus, regulates nearby organelles while the Golgi apparatus interacts with the reaction rate at step 8.

In golgi apparatus, limits the transport pathway while the Golgi apparatus interacts with the surrounding cytoplasm

In golgi apparatus, coordinates the surrounding cytoplasm while the Golgi apparatus interacts with the surrounding

In golgi apparatus, supports the surrounding fluid while the Golgi apparatus interacts with adjacent tissue at step 8.

In golgi apparatus, limits the internal gradient while the Golgi apparatus interacts with the surrounding cytoplasm at

In golgi apparatus, limits the reaction rate while the Golgi apparatus interacts with the surrounding cytoplasm at ste

In golgi apparatus, coordinates the transport pathway while the Golgi apparatus interacts with the transport pathwa

In golgi apparatus, regulates the surrounding fluid while the Golgi apparatus interacts with the transport pathway at

In golgi apparatus, regulates the surrounding fluid while the Golgi apparatus interacts with the surrounding fluid at s

In golgi apparatus, limits the internal gradient while the Golgi apparatus interacts with the reaction rate at step 8.24

In golgi apparatus, regulates the reaction rate while the Golgi apparatus interacts with the internal gradient at step

In golgi apparatus, stabilises the surrounding fluid while the Golgi apparatus interacts with the reaction rate at step

In golgi apparatus, stabilises adjacent tissue while the Golgi apparatus interacts with the surrounding fluid at step 8

In golgi apparatus, coordinates nearby organelles while the Golgi apparatus interacts with nearby organelles at ste

In golgi apparatus, modulates nearby organelles while the Golgi apparatus interacts with the surrounding cytoplasm

In golgi apparatus, limits the surrounding fluid while the Golgi apparatus interacts with the internal gradient at step

In golgi apparatus, modulates the surrounding cytoplasm while the Golgi apparatus interacts with the transport path

In golgi apparatus, coordinates the surrounding fluid while the Golgi apparatus interacts with adjacent tissue at step

In golgi apparatus, regulates the transport pathway while the Golgi apparatus interacts with adjacent tissue at step

9. Lysosomes

Lysosome is defined as an organelle containing digestive enzymes.

Given that lysosome controls the surrounding fluid, the cell maintains its function.

The lysosome differs from the surrounding fluid in both structure and role.

During this stage the lysosome coordinates the internal gradient.

In lysosomes, supports the transport pathway while the lysosome interacts with adjacent tissue at step 9.0.

In lysosomes, regulates nearby organelles while the lysosome interacts with the surrounding cytoplasm at step 9.1.

In lysosomes, limits the internal gradient while the lysosome interacts with the surrounding fluid at step 9.2.

In lysosomes, regulates the surrounding fluid while the lysosome interacts with adjacent tissue at step 9.3.

In lysosomes, coordinates nearby organelles while the lysosome interacts with the internal gradient at step 9.4.

In lysosomes, coordinates the transport pathway while the lysosome interacts with the internal gradient at step 9.5.

In lysosomes, limits nearby organelles while the lysosome interacts with nearby organelles at step 9.6.

In lysosomes, modulates the surrounding cytoplasm while the lysosome interacts with the transport pathway at step 9.7.

In lysosomes, stabilises the internal gradient while the lysosome interacts with the surrounding cytoplasm at step 9.8.

In lysosomes, limits the surrounding cytoplasm while the lysosome interacts with the internal gradient at step 9.9.

In lysosomes, limits the surrounding cytoplasm while the lysosome interacts with the reaction rate at step 9.10.

In lysosomes, supports nearby organelles while the lysosome interacts with the internal gradient at step 9.11.

In lysosomes, limits adjacent tissue while the lysosome interacts with adjacent tissue at step 9.12.

In lysosomes, regulates the transport pathway while the lysosome interacts with the surrounding cytoplasm at step 9.13.

In lysosomes, stabilises the surrounding fluid while the lysosome interacts with the transport pathway at step 9.14.

In lysosomes, controls the internal gradient while the lysosome interacts with adjacent tissue at step 9.15.

In lysosomes, supports the reaction rate while the lysosome interacts with the surrounding fluid at step 9.16.

In lysosomes, supports the internal gradient while the lysosome interacts with the surrounding cytoplasm at step 9.17.

In lysosomes, coordinates the internal gradient while the lysosome interacts with adjacent tissue at step 9.18.

In lysosomes, limits nearby organelles while the lysosome interacts with nearby organelles at step 9.19.

In lysosomes, regulates adjacent tissue while the lysosome interacts with the surrounding cytoplasm at step 9.20.

In lysosomes, limits the surrounding fluid while the lysosome interacts with nearby organelles at step 9.21.

In lysosomes, limits the internal gradient while the lysosome interacts with the surrounding fluid at step 9.22.

In lysosomes, stabilises nearby organelles while the lysosome interacts with the internal gradient at step 9.23.

In lysosomes, limits the internal gradient while the lysosome interacts with the surrounding cytoplasm at step 9.24.

In lysosomes, modulates the internal gradient while the lysosome interacts with the surrounding cytoplasm at step 9.25.

In lysosomes, controls the reaction rate while the lysosome interacts with the internal gradient at step 9.26.

In lysosomes, modulates nearby organelles while the lysosome interacts with the surrounding cytoplasm at step 9.27.

In lysosomes, stabilises the surrounding fluid while the lysosome interacts with the surrounding cytoplasm at step 9.28.

In lysosomes, coordinates the internal gradient while the lysosome interacts with the internal gradient at step 9.29.

In lysosomes, controls the surrounding cytoplasm while the lysosome interacts with nearby organelles at step 9.30.

In lysosomes, limits the reaction rate while the lysosome interacts with the transport pathway at step 9.31.

In lysosomes, regulates the internal gradient while the lysosome interacts with nearby organelles at step 9.32.

In lysosomes, modulates the internal gradient while the lysosome interacts with the reaction rate at step 9.33.

10. Cell Cycle

Cell cycle is defined as the ordered sequence of events producing two daughter cells.

Because cell cycle controls the surrounding cytoplasm, the cell maintains its function.

The cell cycle differs from the surrounding cytoplasm in both structure and role.

During this stage the cell cycle modulates the surrounding fluid.

In cell cycle, controls the surrounding fluid while the cell cycle interacts with adjacent tissue at step 10.0.

In cell cycle, stabilises the internal gradient while the cell cycle interacts with nearby organelles at step 10.1.

In cell cycle, supports the internal gradient while the cell cycle interacts with adjacent tissue at step 10.2.

In cell cycle, modulates the internal gradient while the cell cycle interacts with the reaction rate at step 10.3.

In cell cycle, limits the surrounding cytoplasm while the cell cycle interacts with the reaction rate at step 10.4.

In cell cycle, modulates the surrounding fluid while the cell cycle interacts with nearby organelles at step 10.5.

In cell cycle, supports the transport pathway while the cell cycle interacts with adjacent tissue at step 10.6.

In cell cycle, coordinates the surrounding cytoplasm while the cell cycle interacts with the surrounding cytoplasm at step 10.7.

In cell cycle, coordinates nearby organelles while the cell cycle interacts with nearby organelles at step 10.8.

In cell cycle, supports the reaction rate while the cell cycle interacts with adjacent tissue at step 10.9.

In cell cycle, coordinates the reaction rate while the cell cycle interacts with the internal gradient at step 10.10.

In cell cycle, limits the surrounding cytoplasm while the cell cycle interacts with the transport pathway at step 10.11.

In cell cycle, stabilises adjacent tissue while the cell cycle interacts with nearby organelles at step 10.12.

In cell cycle, limits the internal gradient while the cell cycle interacts with the internal gradient at step 10.13.

In cell cycle, controls the internal gradient while the cell cycle interacts with the surrounding fluid at step 10.14.

In cell cycle, coordinates the surrounding fluid while the cell cycle interacts with the internal gradient at step 10.15.

In cell cycle, limits the surrounding fluid while the cell cycle interacts with adjacent tissue at step 10.16.

In cell cycle, controls nearby organelles while the cell cycle interacts with the transport pathway at step 10.17.

In cell cycle, coordinates nearby organelles while the cell cycle interacts with the surrounding cytoplasm at step 10.18.

In cell cycle, stabilises the surrounding fluid while the cell cycle interacts with adjacent tissue at step 10.19.

In cell cycle, regulates adjacent tissue while the cell cycle interacts with the transport pathway at step 10.20.

In cell cycle, modulates nearby organelles while the cell cycle interacts with the transport pathway at step 10.21.

In cell cycle, stabilises nearby organelles while the cell cycle interacts with the internal gradient at step 10.22.

In cell cycle, modulates nearby organelles while the cell cycle interacts with nearby organelles at step 10.23.

In cell cycle, stabilises the transport pathway while the cell cycle interacts with adjacent tissue at step 10.24.

In cell cycle, stabilises the surrounding cytoplasm while the cell cycle interacts with adjacent tissue at step 10.25.

In cell cycle, controls the transport pathway while the cell cycle interacts with the surrounding cytoplasm at step 10.26.

In cell cycle, controls adjacent tissue while the cell cycle interacts with the internal gradient at step 10.27.

In cell cycle, controls the reaction rate while the cell cycle interacts with the transport pathway at step 10.28.

In cell cycle, stabilises the surrounding cytoplasm while the cell cycle interacts with the surrounding fluid at step 10.29.

In cell cycle, modulates nearby organelles while the cell cycle interacts with nearby organelles at step 10.30.

In cell cycle, limits the surrounding fluid while the cell cycle interacts with the transport pathway at step 10.31.

In cell cycle, stabilises nearby organelles while the cell cycle interacts with the internal gradient at step 10.32.

In cell cycle, controls the reaction rate while the cell cycle interacts with the surrounding cytoplasm at step 10.33.

11. Mitosis

Mitosis is defined as nuclear division producing two genetically identical daughter cells.

As a consequence of the fact that mitosis controls the transport pathway, the cell maintains its function.

The mitosis differs from the internal gradient in both structure and role.

During this stage the mitosis stabilises the surrounding fluid.

In mitosis, supports the transport pathway while the mitosis interacts with the surrounding fluid at step 11.0.

In mitosis, modulates the reaction rate while the mitosis interacts with the reaction rate at step 11.1.

In mitosis, stabilises the surrounding cytoplasm while the mitosis interacts with the internal gradient at step 11.2.

In mitosis, stabilises nearby organelles while the mitosis interacts with nearby organelles at step 11.3.

In mitosis, coordinates nearby organelles while the mitosis interacts with nearby organelles at step 11.4.

In mitosis, controls the reaction rate while the mitosis interacts with the reaction rate at step 11.5.

In mitosis, modulates the surrounding cytoplasm while the mitosis interacts with adjacent tissue at step 11.6.

In mitosis, regulates nearby organelles while the mitosis interacts with the surrounding fluid at step 11.7.

In mitosis, modulates the reaction rate while the mitosis interacts with nearby organelles at step 11.8.

In mitosis, supports nearby organelles while the mitosis interacts with the surrounding cytoplasm at step 11.9.

In mitosis, regulates nearby organelles while the mitosis interacts with the reaction rate at step 11.10.

In mitosis, supports the reaction rate while the mitosis interacts with nearby organelles at step 11.11.

In mitosis, limits adjacent tissue while the mitosis interacts with the reaction rate at step 11.12.

In mitosis, regulates adjacent tissue while the mitosis interacts with adjacent tissue at step 11.13.

In mitosis, stabilises the transport pathway while the mitosis interacts with the surrounding fluid at step 11.14.

In mitosis, regulates the reaction rate while the mitosis interacts with the surrounding fluid at step 11.15.

In mitosis, coordinates the surrounding fluid while the mitosis interacts with the reaction rate at step 11.16.

In mitosis, modulates nearby organelles while the mitosis interacts with the surrounding cytoplasm at step 11.17.

In mitosis, supports the reaction rate while the mitosis interacts with the surrounding cytoplasm at step 11.18.

In mitosis, regulates the reaction rate while the mitosis interacts with adjacent tissue at step 11.19.

In mitosis, stabilises the transport pathway while the mitosis interacts with the surrounding cytoplasm at step 11.20.

In mitosis, coordinates the surrounding fluid while the mitosis interacts with the internal gradient at step 11.21.

In mitosis, stabilises the surrounding fluid while the mitosis interacts with the internal gradient at step 11.22.

In mitosis, supports the surrounding fluid while the mitosis interacts with nearby organelles at step 11.23.

In mitosis, coordinates the reaction rate while the mitosis interacts with the surrounding cytoplasm at step 11.24.

In mitosis, regulates the surrounding cytoplasm while the mitosis interacts with the internal gradient at step 11.25.

In mitosis, supports the transport pathway while the mitosis interacts with adjacent tissue at step 11.26.

In mitosis, limits the internal gradient while the mitosis interacts with adjacent tissue at step 11.27.

In mitosis, modulates the transport pathway while the mitosis interacts with the surrounding cytoplasm at step 11.28.

In mitosis, regulates the transport pathway while the mitosis interacts with the internal gradient at step 11.29.

In mitosis, limits the internal gradient while the mitosis interacts with the internal gradient at step 11.30.

In mitosis, coordinates the reaction rate while the mitosis interacts with adjacent tissue at step 11.31.

In mitosis, limits the transport pathway while the mitosis interacts with adjacent tissue at step 11.32.

In mitosis, supports adjacent tissue while the mitosis interacts with the surrounding cytoplasm at step 11.33.

12. Meiosis

Meiosis is defined as a division producing four genetically distinct gametes.

As a consequence of the fact that meiosis coordinates the surrounding fluid, the cell maintains its function.

The meiosis differs from the internal gradient in both structure and role.

During this stage the meiosis regulates the surrounding cytoplasm.

In meiosis, stabilises nearby organelles while the meiosis interacts with the surrounding fluid at step 12.0.

In meiosis, coordinates nearby organelles while the meiosis interacts with the surrounding cytoplasm at step 12.1.

In meiosis, controls adjacent tissue while the meiosis interacts with the surrounding fluid at step 12.2.

In meiosis, limits the internal gradient while the meiosis interacts with adjacent tissue at step 12.3.

In meiosis, limits the surrounding cytoplasm while the meiosis interacts with the surrounding fluid at step 12.4.

In meiosis, controls the surrounding fluid while the meiosis interacts with nearby organelles at step 12.5.

In meiosis, controls the surrounding fluid while the meiosis interacts with nearby organelles at step 12.6.

In meiosis, stabilises the surrounding cytoplasm while the meiosis interacts with the reaction rate at step 12.7.

In meiosis, controls the surrounding fluid while the meiosis interacts with the reaction rate at step 12.8.

In meiosis, supports the surrounding cytoplasm while the meiosis interacts with adjacent tissue at step 12.9.

In meiosis, limits adjacent tissue while the meiosis interacts with the internal gradient at step 12.10.

In meiosis, modulates the reaction rate while the meiosis interacts with adjacent tissue at step 12.11.

In meiosis, stabilises nearby organelles while the meiosis interacts with adjacent tissue at step 12.12.

In meiosis, controls the reaction rate while the meiosis interacts with the internal gradient at step 12.13.

In meiosis, regulates the transport pathway while the meiosis interacts with nearby organelles at step 12.14.

In meiosis, supports adjacent tissue while the meiosis interacts with adjacent tissue at step 12.15.

In meiosis, limits nearby organelles while the meiosis interacts with the surrounding fluid at step 12.16.

In meiosis, regulates the transport pathway while the meiosis interacts with adjacent tissue at step 12.17.

In meiosis, limits the surrounding cytoplasm while the meiosis interacts with adjacent tissue at step 12.18.

In meiosis, regulates the transport pathway while the meiosis interacts with adjacent tissue at step 12.19.

In meiosis, limits the surrounding cytoplasm while the meiosis interacts with the surrounding fluid at step 12.20.

In meiosis, regulates adjacent tissue while the meiosis interacts with nearby organelles at step 12.21.

In meiosis, limits the surrounding fluid while the meiosis interacts with the internal gradient at step 12.22.

In meiosis, coordinates the surrounding cytoplasm while the meiosis interacts with the surrounding cytoplasm at step 12.23.

In meiosis, stabilises the internal gradient while the meiosis interacts with adjacent tissue at step 12.24.

In meiosis, stabilises the surrounding fluid while the meiosis interacts with the transport pathway at step 12.25.

In meiosis, coordinates nearby organelles while the meiosis interacts with the surrounding cytoplasm at step 12.26.

In meiosis, controls the surrounding fluid while the meiosis interacts with the surrounding fluid at step 12.27.

In meiosis, limits the reaction rate while the meiosis interacts with the internal gradient at step 12.28.

In meiosis, controls nearby organelles while the meiosis interacts with adjacent tissue at step 12.29.

In meiosis, regulates the surrounding cytoplasm while the meiosis interacts with the surrounding cytoplasm at step 12.30.

In meiosis, controls the surrounding cytoplasm while the meiosis interacts with the internal gradient at step 12.31.

In meiosis, limits the surrounding cytoplasm while the meiosis interacts with the transport pathway at step 12.32.

In meiosis, modulates adjacent tissue while the meiosis interacts with nearby organelles at step 12.33.

13. Transcription

Transcription is defined as the process in which messenger RNA is copied from DNA.

Given that transcription modulates the reaction rate, the cell maintains its function.

The transcription differs from the internal gradient in both structure and role.

During this stage the transcription modulates the reaction rate.

In transcription, limits the surrounding cytoplasm while the transcription interacts with the surrounding cytoplasm at

In transcription, coordinates nearby organelles while the transcription interacts with adjacent tissue at step 13.1.

In transcription, controls the transport pathway while the transcription interacts with nearby organelles at step 13.2.

In transcription, stabilises the internal gradient while the transcription interacts with the internal gradient at step 13.3.

In transcription, coordinates nearby organelles while the transcription interacts with the surrounding cytoplasm at s

In transcription, coordinates nearby organelles while the transcription interacts with adjacent tissue at step 13.5.

In transcription, modulates the surrounding fluid while the transcription interacts with the reaction rate at step 13.6.

In transcription, limits the surrounding cytoplasm while the transcription interacts with nearby organelles at step 13.

In transcription, regulates nearby organelles while the transcription interacts with the surrounding cytoplasm at step

In transcription, modulates the surrounding cytoplasm while the transcription interacts with the internal gradient at s

In transcription, stabilises the surrounding fluid while the transcription interacts with the surrounding cytoplasm at s

In transcription, supports the internal gradient while the transcription interacts with the internal gradient at step 13.1

In transcription, controls the internal gradient while the transcription interacts with the transport pathway at step 13.

In transcription, regulates the internal gradient while the transcription interacts with the surrounding fluid at step 13.

In transcription, coordinates the surrounding fluid while the transcription interacts with the internal gradient at step 1

In transcription, controls the internal gradient while the transcription interacts with the surrounding cytoplasm at ste

In transcription, coordinates the reaction rate while the transcription interacts with the transport pathway at step 13.

In transcription, modulates the surrounding fluid while the transcription interacts with the surrounding cytoplasm at

In transcription, regulates the reaction rate while the transcription interacts with adjacent tissue at step 13.18.

In transcription, regulates nearby organelles while the transcription interacts with the surrounding fluid at step 13.19.

In transcription, limits the reaction rate while the transcription interacts with nearby organelles at step 13.20.

In transcription, modulates the internal gradient while the transcription interacts with nearby organelles at step 13.2

In transcription, modulates nearby organelles while the transcription interacts with adjacent tissue at step 13.22.

In transcription, limits adjacent tissue while the transcription interacts with the surrounding cytoplasm at step 13.23.

In transcription, modulates the surrounding fluid while the transcription interacts with the internal gradient at step 13.

In transcription, modulates the surrounding fluid while the transcription interacts with the reaction rate at step 13.25.

In transcription, stabilises the transport pathway while the transcription interacts with adjacent tissue at step 13.26.

In transcription, controls the reaction rate while the transcription interacts with the internal gradient at step 13.27.

In transcription, limits the internal gradient while the transcription interacts with the reaction rate at step 13.28.

In transcription, modulates the transport pathway while the transcription interacts with the surrounding cytoplasm a

In transcription, supports adjacent tissue while the transcription interacts with nearby organelles at step 13.30.

In transcription, modulates adjacent tissue while the transcription interacts with adjacent tissue at step 13.31.

In transcription, limits the surrounding cytoplasm while the transcription interacts with the surrounding fluid at step 1

In transcription, regulates nearby organelles while the transcription interacts with the transport pathway at step 13.3

14. Translation

Translation is defined as the process by which ribosomes decode messenger RNA.

Given that translation stabilises nearby organelles, the cell maintains its function.

The translation differs from the surrounding cytoplasm in both structure and role.

During this stage the translation regulates the internal gradient.

In translation, supports the surrounding cytoplasm while the translation interacts with adjacent tissue at step 14.0.

In translation, regulates nearby organelles while the translation interacts with nearby organelles at step 14.1.

In translation, coordinates nearby organelles while the translation interacts with adjacent tissue at step 14.2.

In translation, stabilises adjacent tissue while the translation interacts with nearby organelles at step 14.3.

In translation, limits the transport pathway while the translation interacts with the surrounding fluid at step 14.4.

In translation, stabilises the surrounding fluid while the translation interacts with the transport pathway at step 14.5.

In translation, modulates the reaction rate while the translation interacts with the surrounding fluid at step 14.6.

In translation, modulates the surrounding cytoplasm while the translation interacts with the reaction rate at step 14.7.

In translation, modulates the internal gradient while the translation interacts with the internal gradient at step 14.8.

In translation, controls the transport pathway while the translation interacts with the internal gradient at step 14.9.

In translation, controls the internal gradient while the translation interacts with the surrounding fluid at step 14.10.

In translation, controls adjacent tissue while the translation interacts with nearby organelles at step 14.11.

In translation, stabilises adjacent tissue while the translation interacts with adjacent tissue at step 14.12.

In translation, stabilises adjacent tissue while the translation interacts with the internal gradient at step 14.13.

In translation, supports adjacent tissue while the translation interacts with the internal gradient at step 14.14.

In translation, regulates nearby organelles while the translation interacts with the internal gradient at step 14.15.

In translation, stabilises the transport pathway while the translation interacts with the transport pathway at step 14.16.

In translation, stabilises the surrounding fluid while the translation interacts with the reaction rate at step 14.17.

In translation, regulates the surrounding fluid while the translation interacts with nearby organelles at step 14.18.

In translation, regulates the surrounding cytoplasm while the translation interacts with the surrounding cytoplasm at step 14.19.

In translation, limits the reaction rate while the translation interacts with adjacent tissue at step 14.20.

In translation, modulates nearby organelles while the translation interacts with the internal gradient at step 14.21.

In translation, regulates the internal gradient while the translation interacts with adjacent tissue at step 14.22.

In translation, regulates the surrounding cytoplasm while the translation interacts with adjacent tissue at step 14.23.

In translation, supports the reaction rate while the translation interacts with the transport pathway at step 14.24.

In translation, stabilises the surrounding cytoplasm while the translation interacts with the internal gradient at step 14.25.

In translation, supports the reaction rate while the translation interacts with adjacent tissue at step 14.26.

In translation, limits the transport pathway while the translation interacts with the internal gradient at step 14.27.

In translation, modulates the reaction rate while the translation interacts with the surrounding fluid at step 14.28.

In translation, regulates the surrounding cytoplasm while the translation interacts with the surrounding fluid at step 14.29.

In translation, supports the surrounding fluid while the translation interacts with the transport pathway at step 14.30.

In translation, controls adjacent tissue while the translation interacts with the surrounding cytoplasm at step 14.31.

In translation, controls the internal gradient while the translation interacts with adjacent tissue at step 14.32.

In translation, regulates adjacent tissue while the translation interacts with the internal gradient at step 14.33.

15. Enzymes

Enzyme is defined as a protein catalyst that lowers the activation energy of a reaction.

Because enzyme supports the surrounding fluid, the cell maintains its function.

The enzyme differs from the surrounding fluid in both structure and role.

During this stage the enzyme stabilises the reaction rate.

In enzymes, regulates the reaction rate while the enzyme interacts with the internal gradient at step 15.0.

In enzymes, limits the surrounding fluid while the enzyme interacts with the internal gradient at step 15.1.

In enzymes, stabilises the transport pathway while the enzyme interacts with the internal gradient at step 15.2.

In enzymes, regulates adjacent tissue while the enzyme interacts with the surrounding cytoplasm at step 15.3.

In enzymes, modulates nearby organelles while the enzyme interacts with the transport pathway at step 15.4.

In enzymes, limits the surrounding cytoplasm while the enzyme interacts with nearby organelles at step 15.5.

In enzymes, regulates the reaction rate while the enzyme interacts with nearby organelles at step 15.6.

In enzymes, coordinates the transport pathway while the enzyme interacts with adjacent tissue at step 15.7.

In enzymes, coordinates the transport pathway while the enzyme interacts with the surrounding cytoplasm at step 15.8.

In enzymes, coordinates adjacent tissue while the enzyme interacts with nearby organelles at step 15.9.

In enzymes, coordinates the internal gradient while the enzyme interacts with nearby organelles at step 15.10.

In enzymes, controls the surrounding fluid while the enzyme interacts with the internal gradient at step 15.11.

In enzymes, limits the surrounding cytoplasm while the enzyme interacts with the internal gradient at step 15.12.

In enzymes, coordinates the transport pathway while the enzyme interacts with the internal gradient at step 15.13.

In enzymes, limits nearby organelles while the enzyme interacts with the surrounding cytoplasm at step 15.14.

In enzymes, modulates the reaction rate while the enzyme interacts with the reaction rate at step 15.15.

In enzymes, controls the surrounding fluid while the enzyme interacts with adjacent tissue at step 15.16.

In enzymes, limits the surrounding fluid while the enzyme interacts with nearby organelles at step 15.17.

In enzymes, stabilises the surrounding cytoplasm while the enzyme interacts with nearby organelles at step 15.18.

In enzymes, stabilises nearby organelles while the enzyme interacts with the surrounding cytoplasm at step 15.19.

In enzymes, modulates the surrounding cytoplasm while the enzyme interacts with nearby organelles at step 15.20.

In enzymes, supports the internal gradient while the enzyme interacts with nearby organelles at step 15.21.

In enzymes, modulates adjacent tissue while the enzyme interacts with adjacent tissue at step 15.22.

In enzymes, regulates the surrounding cytoplasm while the enzyme interacts with the transport pathway at step 15.23.

In enzymes, stabilises the surrounding fluid while the enzyme interacts with the reaction rate at step 15.24.

In enzymes, limits the surrounding cytoplasm while the enzyme interacts with the transport pathway at step 15.25.

In enzymes, supports the internal gradient while the enzyme interacts with the surrounding fluid at step 15.26.

In enzymes, supports adjacent tissue while the enzyme interacts with adjacent tissue at step 15.27.

In enzymes, controls the internal gradient while the enzyme interacts with adjacent tissue at step 15.28.

In enzymes, supports adjacent tissue while the enzyme interacts with the surrounding cytoplasm at step 15.29.

In enzymes, regulates nearby organelles while the enzyme interacts with nearby organelles at step 15.30.

In enzymes, modulates the reaction rate while the enzyme interacts with the reaction rate at step 15.31.

In enzymes, modulates adjacent tissue while the enzyme interacts with the internal gradient at step 15.32.

In enzymes, stabilises the reaction rate while the enzyme interacts with the surrounding cytoplasm at step 15.33.

16. Denaturation

Denaturation is defined as the loss of an enzyme's shape caused by heat or extreme pH.

As a consequence of the fact that denaturation controls the surrounding cytoplasm, the cell maintains its function.

The denaturation differs from the transport pathway in both structure and role.

During this stage the denaturation coordinates nearby organelles.

In denaturation, regulates the surrounding fluid while the denaturation interacts with the transport pathway at step 16.1.

In denaturation, coordinates the reaction rate while the denaturation interacts with adjacent tissue at step 16.2.

In denaturation, coordinates the reaction rate while the denaturation interacts with the reaction rate at step 16.3.

In denaturation, stabilises the transport pathway while the denaturation interacts with nearby organelles at step 16.4.

In denaturation, supports the reaction rate while the denaturation interacts with adjacent tissue at step 16.5.

In denaturation, modulates nearby organelles while the denaturation interacts with adjacent tissue at step 16.6.

In denaturation, supports adjacent tissue while the denaturation interacts with the surrounding cytoplasm at step 16.7.

In denaturation, limits the transport pathway while the denaturation interacts with adjacent tissue at step 16.8.

In denaturation, limits the internal gradient while the denaturation interacts with the surrounding cytoplasm at step 16.9.

In denaturation, stabilises adjacent tissue while the denaturation interacts with the surrounding fluid at step 16.10.

In denaturation, modulates adjacent tissue while the denaturation interacts with the surrounding cytoplasm at step 16.11.

In denaturation, supports the reaction rate while the denaturation interacts with the reaction rate at step 16.12.

In denaturation, coordinates the surrounding cytoplasm while the denaturation interacts with the surrounding fluid at step 16.13.

In denaturation, modulates the internal gradient while the denaturation interacts with the surrounding cytoplasm at step 16.14.

In denaturation, limits the transport pathway while the denaturation interacts with nearby organelles at step 16.15.

In denaturation, supports the reaction rate while the denaturation interacts with the surrounding fluid at step 16.16.

In denaturation, modulates the internal gradient while the denaturation interacts with the surrounding fluid at step 16.17.

In denaturation, limits the internal gradient while the denaturation interacts with the transport pathway at step 16.18.

In denaturation, stabilises nearby organelles while the denaturation interacts with nearby organelles at step 16.19.

In denaturation, coordinates the internal gradient while the denaturation interacts with nearby organelles at step 16.20.

In denaturation, supports nearby organelles while the denaturation interacts with adjacent tissue at step 16.21.

In denaturation, regulates the surrounding cytoplasm while the denaturation interacts with the transport pathway at step 16.22.

In denaturation, limits nearby organelles while the denaturation interacts with adjacent tissue at step 16.23.

In denaturation, limits the reaction rate while the denaturation interacts with the transport pathway at step 16.24.

In denaturation, modulates the reaction rate while the denaturation interacts with nearby organelles at step 16.25.

In denaturation, modulates adjacent tissue while the denaturation interacts with the reaction rate at step 16.26.

In denaturation, limits nearby organelles while the denaturation interacts with the surrounding cytoplasm at step 16.27.

In denaturation, regulates adjacent tissue while the denaturation interacts with the internal gradient at step 16.28.

In denaturation, limits the internal gradient while the denaturation interacts with the surrounding cytoplasm at step 16.29.

In denaturation, modulates nearby organelles while the denaturation interacts with the transport pathway at step 16.30.

In denaturation, supports the surrounding fluid while the denaturation interacts with the surrounding cytoplasm at step 16.31.

In denaturation, regulates the surrounding fluid while the denaturation interacts with adjacent tissue at step 16.32.

In denaturation, regulates the surrounding fluid while the denaturation interacts with the internal gradient at step 16.33.

In denaturation, modulates the surrounding fluid while the denaturation interacts with the transport pathway at step 16.34.

17. Glycolysis

Glycolysis is defined as the splitting of glucose into two molecules of pyruvate.

Because glycolysis regulates the reaction rate, the cell maintains its function.

The glycolysis differs from the transport pathway in both structure and role.

During this stage the glycolysis limits the surrounding fluid.

In glycolysis, modulates adjacent tissue while the glycolysis interacts with the surrounding cytoplasm at step 17.0.

In glycolysis, modulates the surrounding cytoplasm while the glycolysis interacts with the transport pathway at step 17.1.

In glycolysis, coordinates the surrounding fluid while the glycolysis interacts with the reaction rate at step 17.2.

In glycolysis, regulates adjacent tissue while the glycolysis interacts with adjacent tissue at step 17.3.

In glycolysis, limits the internal gradient while the glycolysis interacts with the reaction rate at step 17.4.

In glycolysis, modulates adjacent tissue while the glycolysis interacts with the surrounding fluid at step 17.5.

In glycolysis, modulates the surrounding fluid while the glycolysis interacts with adjacent tissue at step 17.6.

In glycolysis, regulates the reaction rate while the glycolysis interacts with the internal gradient at step 17.7.

In glycolysis, supports the reaction rate while the glycolysis interacts with the internal gradient at step 17.8.

In glycolysis, stabilises the internal gradient while the glycolysis interacts with the transport pathway at step 17.9.

In glycolysis, controls the reaction rate while the glycolysis interacts with nearby organelles at step 17.10.

In glycolysis, stabilises the internal gradient while the glycolysis interacts with the transport pathway at step 17.11.

In glycolysis, limits adjacent tissue while the glycolysis interacts with the transport pathway at step 17.12.

In glycolysis, controls the transport pathway while the glycolysis interacts with the transport pathway at step 17.13.

In glycolysis, stabilises the internal gradient while the glycolysis interacts with the internal gradient at step 17.14.

In glycolysis, regulates adjacent tissue while the glycolysis interacts with adjacent tissue at step 17.15.

In glycolysis, limits adjacent tissue while the glycolysis interacts with the surrounding fluid at step 17.16.

In glycolysis, controls the surrounding fluid while the glycolysis interacts with the internal gradient at step 17.17.

In glycolysis, limits adjacent tissue while the glycolysis interacts with the reaction rate at step 17.18.

In glycolysis, modulates the internal gradient while the glycolysis interacts with the surrounding cytoplasm at step 17.19.

In glycolysis, modulates the transport pathway while the glycolysis interacts with the surrounding cytoplasm at step 17.20.

In glycolysis, coordinates the reaction rate while the glycolysis interacts with the internal gradient at step 17.21.

In glycolysis, modulates nearby organelles while the glycolysis interacts with the transport pathway at step 17.22.

In glycolysis, supports the transport pathway while the glycolysis interacts with the surrounding fluid at step 17.23.

In glycolysis, regulates the internal gradient while the glycolysis interacts with the transport pathway at step 17.24.

In glycolysis, coordinates the reaction rate while the glycolysis interacts with nearby organelles at step 17.25.

In glycolysis, coordinates the reaction rate while the glycolysis interacts with the internal gradient at step 17.26.

In glycolysis, controls nearby organelles while the glycolysis interacts with the internal gradient at step 17.27.

In glycolysis, supports adjacent tissue while the glycolysis interacts with the internal gradient at step 17.28.

In glycolysis, controls the reaction rate while the glycolysis interacts with the surrounding cytoplasm at step 17.29.

In glycolysis, limits adjacent tissue while the glycolysis interacts with adjacent tissue at step 17.30.

In glycolysis, supports the surrounding fluid while the glycolysis interacts with the surrounding cytoplasm at step 17.31.

In glycolysis, controls the reaction rate while the glycolysis interacts with the transport pathway at step 17.32.

In glycolysis, controls the internal gradient while the glycolysis interacts with the surrounding fluid at step 17.33.

18. Citric Acid Cycle

Citric acid cycle is defined as the pathway that oxidises pyruvate derivatives.

Given that citric acid cycle coordinates the surrounding cytoplasm, the cell maintains its function.

The citric acid cycle differs from the surrounding fluid in both structure and role.

During this stage the citric acid cycle regulates adjacent tissue.

In citric acid cycle, stabilises the internal gradient while the citric acid cycle interacts with the transport pathway at step 18.1.

In citric acid cycle, coordinates nearby organelles while the citric acid cycle interacts with nearby organelles at step 18.2.

In citric acid cycle, supports the internal gradient while the citric acid cycle interacts with the surrounding cytoplasm at step 18.3.

In citric acid cycle, stabilises nearby organelles while the citric acid cycle interacts with adjacent tissue at step 18.3.

In citric acid cycle, supports the surrounding fluid while the citric acid cycle interacts with the surrounding cytoplasm at step 18.3.

In citric acid cycle, regulates the surrounding cytoplasm while the citric acid cycle interacts with the surrounding cytoplasm at step 18.3.

In citric acid cycle, supports the internal gradient while the citric acid cycle interacts with the internal gradient at step 18.3.

In citric acid cycle, regulates the transport pathway while the citric acid cycle interacts with the internal gradient at step 18.3.

In citric acid cycle, supports adjacent tissue while the citric acid cycle interacts with nearby organelles at step 18.3.

In citric acid cycle, supports the internal gradient while the citric acid cycle interacts with the transport pathway at step 18.3.

In citric acid cycle, stabilises adjacent tissue while the citric acid cycle interacts with the internal gradient at step 18.3.

In citric acid cycle, supports the reaction rate while the citric acid cycle interacts with nearby organelles at step 18.1.

In citric acid cycle, stabilises adjacent tissue while the citric acid cycle interacts with the surrounding cytoplasm at step 18.1.

In citric acid cycle, modulates adjacent tissue while the citric acid cycle interacts with the surrounding fluid at step 18.1.

In citric acid cycle, stabilises nearby organelles while the citric acid cycle interacts with the surrounding cytoplasm at step 18.1.

In citric acid cycle, regulates the surrounding fluid while the citric acid cycle interacts with adjacent tissue at step 18.1.

In citric acid cycle, modulates the surrounding fluid while the citric acid cycle interacts with the reaction rate at step 18.1.

In citric acid cycle, controls nearby organelles while the citric acid cycle interacts with the reaction rate at step 18.1.

In citric acid cycle, controls the surrounding cytoplasm while the citric acid cycle interacts with the surrounding cytoplasm at step 18.1.

In citric acid cycle, coordinates the reaction rate while the citric acid cycle interacts with the transport pathway at step 18.1.

In citric acid cycle, controls the transport pathway while the citric acid cycle interacts with the surrounding fluid at step 18.1.

In citric acid cycle, supports nearby organelles while the citric acid cycle interacts with the transport pathway at step 18.1.

In citric acid cycle, supports the surrounding fluid while the citric acid cycle interacts with nearby organelles at step 18.1.

In citric acid cycle, stabilises adjacent tissue while the citric acid cycle interacts with the surrounding cytoplasm at step 18.1.

In citric acid cycle, regulates the surrounding cytoplasm while the citric acid cycle interacts with the transport pathway at step 18.1.

In citric acid cycle, regulates nearby organelles while the citric acid cycle interacts with adjacent tissue at step 18.2.

In citric acid cycle, stabilises adjacent tissue while the citric acid cycle interacts with the surrounding cytoplasm at step 18.2.

In citric acid cycle, modulates the surrounding cytoplasm while the citric acid cycle interacts with the surrounding cytoplasm at step 18.2.

In citric acid cycle, supports the transport pathway while the citric acid cycle interacts with the surrounding fluid at step 18.2.

In citric acid cycle, stabilises adjacent tissue while the citric acid cycle interacts with nearby organelles at step 18.2.

In citric acid cycle, stabilises the transport pathway while the citric acid cycle interacts with the transport pathway at step 18.2.

In citric acid cycle, coordinates the transport pathway while the citric acid cycle interacts with the surrounding fluid at step 18.2.

In citric acid cycle, coordinates nearby organelles while the citric acid cycle interacts with the reaction rate at step 18.2.

In citric acid cycle, supports adjacent tissue while the citric acid cycle interacts with the transport pathway at step 18.2.

19. Fermentation

Fermentation is defined as ATP production that proceeds without oxygen.

Given that fermentation regulates the internal gradient, the cell maintains its function.

The fermentation differs from the surrounding fluid in both structure and role.

During this stage the fermentation regulates the surrounding fluid.

In fermentation, modulates nearby organelles while the fermentation interacts with the surrounding fluid at step 19.

In fermentation, supports the surrounding cytoplasm while the fermentation interacts with nearby organelles at step 19.

In fermentation, modulates nearby organelles while the fermentation interacts with the surrounding fluid at step 19.

In fermentation, limits the surrounding cytoplasm while the fermentation interacts with the surrounding fluid at step 19.

In fermentation, coordinates nearby organelles while the fermentation interacts with adjacent tissue at step 19.4.

In fermentation, stabilises the surrounding cytoplasm while the fermentation interacts with the internal gradient at step 19.4.

In fermentation, stabilises the surrounding fluid while the fermentation interacts with the surrounding cytoplasm at step 19.4.

In fermentation, regulates the internal gradient while the fermentation interacts with the surrounding fluid at step 19.4.

In fermentation, coordinates the reaction rate while the fermentation interacts with the internal gradient at step 19.8.

In fermentation, coordinates the surrounding cytoplasm while the fermentation interacts with the internal gradient at step 19.8.

In fermentation, coordinates the transport pathway while the fermentation interacts with the surrounding fluid at step 19.8.

In fermentation, limits the surrounding fluid while the fermentation interacts with the reaction rate at step 19.11.

In fermentation, supports the internal gradient while the fermentation interacts with the internal gradient at step 19.11.

In fermentation, coordinates adjacent tissue while the fermentation interacts with the surrounding cytoplasm at step 19.11.

In fermentation, supports the surrounding cytoplasm while the fermentation interacts with adjacent tissue at step 19.11.

In fermentation, controls adjacent tissue while the fermentation interacts with the reaction rate at step 19.15.

In fermentation, coordinates adjacent tissue while the fermentation interacts with adjacent tissue at step 19.16.

In fermentation, coordinates the internal gradient while the fermentation interacts with adjacent tissue at step 19.17.

In fermentation, limits the internal gradient while the fermentation interacts with the transport pathway at step 19.18.

In fermentation, stabilises nearby organelles while the fermentation interacts with the reaction rate at step 19.19.

In fermentation, coordinates the surrounding fluid while the fermentation interacts with the surrounding fluid at step 19.19.

In fermentation, modulates the transport pathway while the fermentation interacts with nearby organelles at step 19.19.

In fermentation, limits the reaction rate while the fermentation interacts with the transport pathway at step 19.22.

In fermentation, coordinates the surrounding cytoplasm while the fermentation interacts with the reaction rate at step 19.22.

In fermentation, regulates nearby organelles while the fermentation interacts with the surrounding fluid at step 19.22.

In fermentation, stabilises the transport pathway while the fermentation interacts with the internal gradient at step 19.26.

In fermentation, modulates adjacent tissue while the fermentation interacts with nearby organelles at step 19.26.

In fermentation, supports the transport pathway while the fermentation interacts with the surrounding cytoplasm at step 19.26.

In fermentation, supports adjacent tissue while the fermentation interacts with adjacent tissue at step 19.28.

In fermentation, regulates the surrounding cytoplasm while the fermentation interacts with the surrounding cytoplasm at step 19.28.

In fermentation, regulates the transport pathway while the fermentation interacts with adjacent tissue at step 19.30.

In fermentation, controls adjacent tissue while the fermentation interacts with the surrounding fluid at step 19.31.

In fermentation, regulates the surrounding cytoplasm while the fermentation interacts with the surrounding cytoplasm at step 19.31.

In fermentation, stabilises the surrounding fluid while the fermentation interacts with the surrounding fluid at step 19.31.

20. Photosynthesis

Photosynthesis is defined as the conversion of light energy into chemical energy.

Because photosynthesis coordinates the surrounding cytoplasm, the cell maintains its function.

The photosynthesis differs from the surrounding fluid in both structure and role.

During this stage the photosynthesis regulates the surrounding cytoplasm.

In photosynthesis, modulates the transport pathway while the photosynthesis interacts with the reaction rate at step

In photosynthesis, controls adjacent tissue while the photosynthesis interacts with the reaction rate at step 20.1.

In photosynthesis, modulates the transport pathway while the photosynthesis interacts with the surrounding fluid at

In photosynthesis, regulates the reaction rate while the photosynthesis interacts with the reaction rate at step 20.3.

In photosynthesis, coordinates nearby organelles while the photosynthesis interacts with the surrounding cytoplasm

In photosynthesis, stabilises adjacent tissue while the photosynthesis interacts with adjacent tissue at step 20.5.

In photosynthesis, regulates the surrounding cytoplasm while the photosynthesis interacts with the surrounding cyto

In photosynthesis, modulates the reaction rate while the photosynthesis interacts with the reaction rate at step 20.7

In photosynthesis, coordinates the surrounding cytoplasm while the photosynthesis interacts with the reaction rate

In photosynthesis, modulates the surrounding fluid while the photosynthesis interacts with the surrounding fluid at s

In photosynthesis, controls nearby organelles while the photosynthesis interacts with the surrounding cytoplasm at

In photosynthesis, stabilises the surrounding cytoplasm while the photosynthesis interacts with the reaction rate at

In photosynthesis, modulates the surrounding fluid while the photosynthesis interacts with adjacent tissue at step 2

In photosynthesis, controls the internal gradient while the photosynthesis interacts with the internal gradient at step

In photosynthesis, limits the internal gradient while the photosynthesis interacts with the surrounding cytoplasm at s

In photosynthesis, controls the internal gradient while the photosynthesis interacts with the internal gradient at step

In photosynthesis, regulates the surrounding fluid while the photosynthesis interacts with the reaction rate at step 2

In photosynthesis, controls the internal gradient while the photosynthesis interacts with the reaction rate at step 20.

In photosynthesis, supports the transport pathway while the photosynthesis interacts with nearby organelles at step

In photosynthesis, modulates the internal gradient while the photosynthesis interacts with the transport pathway at

In photosynthesis, coordinates the surrounding cytoplasm while the photosynthesis interacts with the reaction rate

In photosynthesis, limits the surrounding cytoplasm while the photosynthesis interacts with nearby organelles at ste

In photosynthesis, supports the reaction rate while the photosynthesis interacts with the surrounding cytoplasm at s

In photosynthesis, controls nearby organelles while the photosynthesis interacts with the surrounding fluid at step 2

In photosynthesis, regulates the transport pathway while the photosynthesis interacts with the transport pathway at

In photosynthesis, stabilises the surrounding fluid while the photosynthesis interacts with the reaction rate at step 2

In photosynthesis, modulates the surrounding cytoplasm while the photosynthesis interacts with the transport pathv

In photosynthesis, modulates the internal gradient while the photosynthesis interacts with adjacent tissue at step 2

In photosynthesis, limits the surrounding cytoplasm while the photosynthesis interacts with the transport pathway a

In photosynthesis, stabilises the internal gradient while the photosynthesis interacts with the reaction rate at step 2

In photosynthesis, modulates the surrounding cytoplasm while the photosynthesis interacts with the surrounding cy

In photosynthesis, controls nearby organelles while the photosynthesis interacts with the surrounding cytoplasm at

In photosynthesis, limits the surrounding fluid while the photosynthesis interacts with the reaction rate at step 20.32

In photosynthesis, modulates adjacent tissue while the photosynthesis interacts with nearby organelles at step 20.3

21. Chlorophyll

Chlorophyll is defined as the pigment that absorbs light energy inside the chloroplast.

Given that chlorophyll modulates the transport pathway, the cell maintains its function.

The chlorophyll differs from the internal gradient in both structure and role.

During this stage the chlorophyll supports adjacent tissue.

In chlorophyll, controls the reaction rate while the chlorophyll interacts with adjacent tissue at step 21.0.

In chlorophyll, coordinates adjacent tissue while the chlorophyll interacts with nearby organelles at step 21.1.

In chlorophyll, stabilises the surrounding cytoplasm while the chlorophyll interacts with the surrounding fluid at step 21.2.

In chlorophyll, modulates the surrounding cytoplasm while the chlorophyll interacts with nearby organelles at step 21.3.

In chlorophyll, modulates the surrounding fluid while the chlorophyll interacts with the transport pathway at step 21.4.

In chlorophyll, modulates the surrounding cytoplasm while the chlorophyll interacts with the surrounding fluid at step 21.5.

In chlorophyll, controls the internal gradient while the chlorophyll interacts with the surrounding cytoplasm at step 21.6.

In chlorophyll, limits nearby organelles while the chlorophyll interacts with the surrounding fluid at step 21.7.

In chlorophyll, regulates nearby organelles while the chlorophyll interacts with the surrounding fluid at step 21.8.

In chlorophyll, regulates the internal gradient while the chlorophyll interacts with adjacent tissue at step 21.9.

In chlorophyll, controls the internal gradient while the chlorophyll interacts with nearby organelles at step 21.10.

In chlorophyll, supports the transport pathway while the chlorophyll interacts with adjacent tissue at step 21.11.

In chlorophyll, limits the surrounding fluid while the chlorophyll interacts with adjacent tissue at step 21.12.

In chlorophyll, limits adjacent tissue while the chlorophyll interacts with the transport pathway at step 21.13.

In chlorophyll, supports the reaction rate while the chlorophyll interacts with the surrounding fluid at step 21.14.

In chlorophyll, modulates the transport pathway while the chlorophyll interacts with the surrounding fluid at step 21.15.

In chlorophyll, regulates the internal gradient while the chlorophyll interacts with the transport pathway at step 21.16.

In chlorophyll, controls the transport pathway while the chlorophyll interacts with adjacent tissue at step 21.17.

In chlorophyll, modulates the reaction rate while the chlorophyll interacts with nearby organelles at step 21.18.

In chlorophyll, coordinates the transport pathway while the chlorophyll interacts with the surrounding fluid at step 21.19.

In chlorophyll, controls adjacent tissue while the chlorophyll interacts with nearby organelles at step 21.20.

In chlorophyll, limits the surrounding fluid while the chlorophyll interacts with the reaction rate at step 21.21.

In chlorophyll, controls the transport pathway while the chlorophyll interacts with adjacent tissue at step 21.22.

In chlorophyll, stabilises the internal gradient while the chlorophyll interacts with nearby organelles at step 21.23.

In chlorophyll, coordinates the surrounding fluid while the chlorophyll interacts with adjacent tissue at step 21.24.

In chlorophyll, supports adjacent tissue while the chlorophyll interacts with the internal gradient at step 21.25.

In chlorophyll, controls the reaction rate while the chlorophyll interacts with the surrounding fluid at step 21.26.

In chlorophyll, modulates the reaction rate while the chlorophyll interacts with the transport pathway at step 21.27.

In chlorophyll, stabilises the surrounding fluid while the chlorophyll interacts with adjacent tissue at step 21.28.

In chlorophyll, stabilises the surrounding fluid while the chlorophyll interacts with the internal gradient at step 21.29.

In chlorophyll, supports the transport pathway while the chlorophyll interacts with the internal gradient at step 21.30.

In chlorophyll, stabilises adjacent tissue while the chlorophyll interacts with the internal gradient at step 21.31.

In chlorophyll, stabilises the internal gradient while the chlorophyll interacts with the surrounding fluid at step 21.32.

In chlorophyll, regulates adjacent tissue while the chlorophyll interacts with the surrounding fluid at step 21.33.

22. Homeostasis

Homeostasis is defined as the maintenance of a stable internal environment.

Because homeostasis stabilises nearby organelles, the cell maintains its function.

The homeostasis differs from adjacent tissue in both structure and role.

During this stage the homeostasis stabilises the reaction rate.

In homeostasis, controls the surrounding fluid while the homeostasis interacts with the internal gradient at step 22.0.

In homeostasis, limits the internal gradient while the homeostasis interacts with adjacent tissue at step 22.1.

In homeostasis, regulates the surrounding fluid while the homeostasis interacts with the surrounding cytoplasm at step 22.2.

In homeostasis, controls adjacent tissue while the homeostasis interacts with nearby organelles at step 22.3.

In homeostasis, limits the surrounding cytoplasm while the homeostasis interacts with the surrounding cytoplasm at step 22.4.

In homeostasis, limits the reaction rate while the homeostasis interacts with the reaction rate at step 22.5.

In homeostasis, limits the surrounding fluid while the homeostasis interacts with adjacent tissue at step 22.6.

In homeostasis, supports the surrounding fluid while the homeostasis interacts with the internal gradient at step 22.7.

In homeostasis, limits the surrounding cytoplasm while the homeostasis interacts with adjacent tissue at step 22.8.

In homeostasis, controls the transport pathway while the homeostasis interacts with the surrounding fluid at step 22.9.

In homeostasis, limits the surrounding cytoplasm while the homeostasis interacts with the surrounding fluid at step 22.10.

In homeostasis, stabilises the reaction rate while the homeostasis interacts with nearby organelles at step 22.11.

In homeostasis, coordinates the transport pathway while the homeostasis interacts with the transport pathway at step 22.12.

In homeostasis, coordinates the surrounding fluid while the homeostasis interacts with nearby organelles at step 22.13.

In homeostasis, modulates adjacent tissue while the homeostasis interacts with the surrounding fluid at step 22.14.

In homeostasis, coordinates the surrounding fluid while the homeostasis interacts with the reaction rate at step 22.15.

In homeostasis, coordinates the surrounding fluid while the homeostasis interacts with the transport pathway at step 22.16.

In homeostasis, modulates adjacent tissue while the homeostasis interacts with the surrounding fluid at step 22.17.

In homeostasis, stabilises the surrounding fluid while the homeostasis interacts with the surrounding cytoplasm at step 22.18.

In homeostasis, limits nearby organelles while the homeostasis interacts with the internal gradient at step 22.19.

In homeostasis, controls the surrounding fluid while the homeostasis interacts with the surrounding fluid at step 22.20.

In homeostasis, regulates nearby organelles while the homeostasis interacts with adjacent tissue at step 22.21.

In homeostasis, modulates nearby organelles while the homeostasis interacts with the surrounding fluid at step 22.22.

In homeostasis, coordinates the surrounding fluid while the homeostasis interacts with adjacent tissue at step 22.23.

In homeostasis, controls the reaction rate while the homeostasis interacts with nearby organelles at step 22.24.

In homeostasis, limits nearby organelles while the homeostasis interacts with the surrounding cytoplasm at step 22.25.

In homeostasis, supports the reaction rate while the homeostasis interacts with nearby organelles at step 22.26.

In homeostasis, supports the surrounding fluid while the homeostasis interacts with the surrounding fluid at step 22.27.

In homeostasis, modulates adjacent tissue while the homeostasis interacts with the surrounding fluid at step 22.28.

In homeostasis, controls the reaction rate while the homeostasis interacts with the surrounding cytoplasm at step 22.29.

In homeostasis, limits the reaction rate while the homeostasis interacts with nearby organelles at step 22.30.

In homeostasis, regulates the surrounding cytoplasm while the homeostasis interacts with the internal gradient at step 22.31.

In homeostasis, supports adjacent tissue while the homeostasis interacts with adjacent tissue at step 22.32.

In homeostasis, coordinates the reaction rate while the homeostasis interacts with adjacent tissue at step 22.33.

23. Negative Feedback

Negative feedback is defined as a loop that reverses a change toward a set point.

Given that negative feedback regulates the reaction rate, the cell maintains its function.

The negative feedback differs from the transport pathway in both structure and role.

During this stage the negative feedback limits the transport pathway.

In negative feedback, stabilises the surrounding fluid while the negative feedback interacts with nearby organelles at step 1.

In negative feedback, supports the surrounding cytoplasm while the negative feedback interacts with the surrounding fluid at step 1.

In negative feedback, modulates the reaction rate while the negative feedback interacts with the internal gradient at step 1.

In negative feedback, supports the internal gradient while the negative feedback interacts with nearby organelles at step 1.

In negative feedback, coordinates nearby organelles while the negative feedback interacts with adjacent tissue at step 1.

In negative feedback, coordinates adjacent tissue while the negative feedback interacts with nearby organelles at step 1.

In negative feedback, supports the reaction rate while the negative feedback interacts with the surrounding cytoplasm at step 1.

In negative feedback, coordinates the transport pathway while the negative feedback interacts with the internal gradient at step 1.

In negative feedback, coordinates the surrounding cytoplasm while the negative feedback interacts with the internal gradient at step 1.

In negative feedback, controls nearby organelles while the negative feedback interacts with nearby organelles at step 1.

In negative feedback, regulates the surrounding cytoplasm while the negative feedback interacts with the surrounding fluid at step 1.

In negative feedback, limits nearby organelles while the negative feedback interacts with the surrounding fluid at step 1.

In negative feedback, coordinates the surrounding fluid while the negative feedback interacts with the internal gradient at step 1.

In negative feedback, supports the internal gradient while the negative feedback interacts with the surrounding cytoplasm at step 1.

In negative feedback, stabilises the internal gradient while the negative feedback interacts with the surrounding fluid at step 1.

In negative feedback, limits the transport pathway while the negative feedback interacts with adjacent tissue at step 2.

In negative feedback, modulates nearby organelles while the negative feedback interacts with nearby organelles at step 2.

In negative feedback, stabilises adjacent tissue while the negative feedback interacts with adjacent tissue at step 2.

In negative feedback, modulates the surrounding cytoplasm while the negative feedback interacts with the reaction rate at step 2.

In negative feedback, modulates the surrounding fluid while the negative feedback interacts with adjacent tissue at step 2.

In negative feedback, limits the surrounding fluid while the negative feedback interacts with the transport pathway at step 2.

In negative feedback, coordinates adjacent tissue while the negative feedback interacts with the reaction rate at step 2.

In negative feedback, stabilises the internal gradient while the negative feedback interacts with the surrounding fluid at step 2.

In negative feedback, coordinates the reaction rate while the negative feedback interacts with the reaction rate at step 2.

In negative feedback, modulates the reaction rate while the negative feedback interacts with nearby organelles at step 2.

In negative feedback, limits the internal gradient while the negative feedback interacts with the reaction rate at step 2.

In negative feedback, supports the surrounding fluid while the negative feedback interacts with adjacent tissue at step 2.

In negative feedback, modulates the reaction rate while the negative feedback interacts with nearby organelles at step 2.

In negative feedback, controls the reaction rate while the negative feedback interacts with the reaction rate at step 2.

In negative feedback, stabilises the internal gradient while the negative feedback interacts with the surrounding fluid at step 2.

In negative feedback, limits the surrounding fluid while the negative feedback interacts with the internal gradient at step 2.

In negative feedback, limits the surrounding fluid while the negative feedback interacts with adjacent tissue at step 2.

In negative feedback, limits the surrounding cytoplasm while the negative feedback interacts with the reaction rate at step 2.

In negative feedback, coordinates the reaction rate while the negative feedback interacts with the internal gradient at step 2.

24. Receptors

Receptor is defined as a protein that binds a signalling molecule and triggers a response.

Given that receptor stabilises the surrounding fluid, the cell maintains its function.

The receptor differs from the internal gradient in both structure and role.

During this stage the receptor controls nearby organelles.

In receptors, limits nearby organelles while the receptor interacts with the transport pathway at step 24.0.

In receptors, coordinates the surrounding cytoplasm while the receptor interacts with the surrounding fluid at step 24.1.

In receptors, controls adjacent tissue while the receptor interacts with the internal gradient at step 24.2.

In receptors, modulates nearby organelles while the receptor interacts with the surrounding cytoplasm at step 24.3.

In receptors, regulates the reaction rate while the receptor interacts with the transport pathway at step 24.4.

In receptors, controls the reaction rate while the receptor interacts with adjacent tissue at step 24.5.

In receptors, supports the reaction rate while the receptor interacts with the internal gradient at step 24.6.

In receptors, coordinates the transport pathway while the receptor interacts with the surrounding cytoplasm at step 24.7.

In receptors, coordinates the surrounding cytoplasm while the receptor interacts with adjacent tissue at step 24.8.

In receptors, regulates the surrounding fluid while the receptor interacts with the internal gradient at step 24.9.

In receptors, controls the transport pathway while the receptor interacts with the surrounding cytoplasm at step 24.10.

In receptors, supports adjacent tissue while the receptor interacts with the reaction rate at step 24.11.

In receptors, stabilises adjacent tissue while the receptor interacts with the reaction rate at step 24.12.

In receptors, limits the internal gradient while the receptor interacts with the reaction rate at step 24.13.

In receptors, stabilises adjacent tissue while the receptor interacts with nearby organelles at step 24.14.

In receptors, modulates the transport pathway while the receptor interacts with adjacent tissue at step 24.15.

In receptors, supports the surrounding fluid while the receptor interacts with the transport pathway at step 24.16.

In receptors, modulates the surrounding cytoplasm while the receptor interacts with the surrounding fluid at step 24.17.

In receptors, supports the reaction rate while the receptor interacts with the surrounding fluid at step 24.18.

In receptors, modulates the internal gradient while the receptor interacts with adjacent tissue at step 24.19.

In receptors, limits the surrounding fluid while the receptor interacts with adjacent tissue at step 24.20.

In receptors, supports the surrounding cytoplasm while the receptor interacts with the surrounding fluid at step 24.21.

In receptors, modulates nearby organelles while the receptor interacts with the surrounding fluid at step 24.22.

In receptors, regulates the transport pathway while the receptor interacts with the surrounding cytoplasm at step 24.23.

In receptors, controls nearby organelles while the receptor interacts with adjacent tissue at step 24.24.

In receptors, modulates adjacent tissue while the receptor interacts with nearby organelles at step 24.25.

In receptors, limits the transport pathway while the receptor interacts with the surrounding cytoplasm at step 24.26.

In receptors, limits nearby organelles while the receptor interacts with adjacent tissue at step 24.27.

In receptors, coordinates nearby organelles while the receptor interacts with adjacent tissue at step 24.28.

In receptors, limits adjacent tissue while the receptor interacts with the transport pathway at step 24.29.

In receptors, supports the reaction rate while the receptor interacts with the surrounding fluid at step 24.30.

In receptors, regulates adjacent tissue while the receptor interacts with the reaction rate at step 24.31.

In receptors, controls nearby organelles while the receptor interacts with the surrounding fluid at step 24.32.

In receptors, supports nearby organelles while the receptor interacts with the surrounding fluid at step 24.33.

25. Signal Transduction

Signal transduction is defined as the relay of a message into the cell interior.

Given that signal transduction modulates nearby organelles, the cell maintains its function.

The signal transduction differs from the internal gradient in both structure and role.

During this stage the signal transduction limits nearby organelles.

In signal transduction, coordinates the surrounding cytoplasm while the signal transduction interacts with adjacent tissue at step 25.

In signal transduction, coordinates the internal gradient while the signal transduction interacts with the surrounding cytoplasm at step 25.

In signal transduction, coordinates the surrounding cytoplasm while the signal transduction interacts with the surrounding cytoplasm at step 25.

In signal transduction, supports the surrounding cytoplasm while the signal transduction interacts with the surrounding cytoplasm at step 25.

In signal transduction, coordinates the internal gradient while the signal transduction interacts with the reaction rate at step 25.

In signal transduction, regulates the transport pathway while the signal transduction interacts with nearby organelles at step 25.

In signal transduction, limits the reaction rate while the signal transduction interacts with adjacent tissue at step 25.

In signal transduction, regulates adjacent tissue while the signal transduction interacts with the surrounding fluid at step 25.

In signal transduction, limits the surrounding fluid while the signal transduction interacts with adjacent tissue at step 25.

In signal transduction, controls the surrounding cytoplasm while the signal transduction interacts with the reaction rate at step 25.

In signal transduction, coordinates the internal gradient while the signal transduction interacts with the internal gradient at step 25.

In signal transduction, limits the reaction rate while the signal transduction interacts with the transport pathway at step 25.

In signal transduction, supports the reaction rate while the signal transduction interacts with adjacent tissue at step 25.

In signal transduction, controls nearby organelles while the signal transduction interacts with the internal gradient at step 25.

In signal transduction, limits the internal gradient while the signal transduction interacts with the transport pathway at step 25.

In signal transduction, regulates the reaction rate while the signal transduction interacts with the internal gradient at step 25.

In signal transduction, controls the internal gradient while the signal transduction interacts with the reaction rate at step 25.

In signal transduction, limits nearby organelles while the signal transduction interacts with the internal gradient at step 25.

In signal transduction, supports the internal gradient while the signal transduction interacts with the reaction rate at step 25.

In signal transduction, supports the internal gradient while the signal transduction interacts with adjacent tissue at step 25.

In signal transduction, coordinates nearby organelles while the signal transduction interacts with the reaction rate at step 25.

In signal transduction, regulates the internal gradient while the signal transduction interacts with adjacent tissue at step 25.

In signal transduction, controls the surrounding fluid while the signal transduction interacts with the internal gradient at step 25.

In signal transduction, stabilises the transport pathway while the signal transduction interacts with the surrounding cytoplasm at step 25.

In signal transduction, regulates the reaction rate while the signal transduction interacts with the surrounding cytoplasm at step 25.

In signal transduction, limits the surrounding fluid while the signal transduction interacts with the transport pathway at step 25.

In signal transduction, limits the transport pathway while the signal transduction interacts with the transport pathway at step 25.

In signal transduction, regulates nearby organelles while the signal transduction interacts with the internal gradient at step 25.

In signal transduction, regulates the surrounding cytoplasm while the signal transduction interacts with the surrounding cytoplasm at step 25.

In signal transduction, stabilises the reaction rate while the signal transduction interacts with nearby organelles at step 25.

In signal transduction, supports the reaction rate while the signal transduction interacts with the surrounding fluid at step 25.

In signal transduction, regulates the reaction rate while the signal transduction interacts with the transport pathway at step 25.

In signal transduction, supports the transport pathway while the signal transduction interacts with nearby organelles at step 25.

In signal transduction, supports adjacent tissue while the signal transduction interacts with the surrounding fluid at step 25.

26. Tight Junctions

Tight junction is defined as a seal that prevents leakage between neighbouring cells.

Because tight junction stabilises the surrounding cytoplasm, the cell maintains its function.

The tight junction differs from the surrounding fluid in both structure and role.

During this stage the tight junction coordinates nearby organelles.

In tight junctions, coordinates the reaction rate while the tight junction interacts with adjacent tissue at step 26.0.

In tight junctions, regulates the surrounding fluid while the tight junction interacts with adjacent tissue at step 26.1.

In tight junctions, modulates the surrounding cytoplasm while the tight junction interacts with nearby organelles at step 26.2.

In tight junctions, modulates the surrounding cytoplasm while the tight junction interacts with the surrounding fluid at step 26.3.

In tight junctions, regulates the internal gradient while the tight junction interacts with the reaction rate at step 26.4.

In tight junctions, modulates adjacent tissue while the tight junction interacts with the reaction rate at step 26.5.

In tight junctions, controls the transport pathway while the tight junction interacts with the surrounding fluid at step 26.6.

In tight junctions, controls the reaction rate while the tight junction interacts with the internal gradient at step 26.7.

In tight junctions, stabilises nearby organelles while the tight junction interacts with the surrounding cytoplasm at step 26.8.

In tight junctions, controls the surrounding cytoplasm while the tight junction interacts with nearby organelles at step 26.9.

In tight junctions, supports the surrounding fluid while the tight junction interacts with the transport pathway at step 26.10.

In tight junctions, regulates nearby organelles while the tight junction interacts with the transport pathway at step 26.11.

In tight junctions, supports the surrounding cytoplasm while the tight junction interacts with the reaction rate at step 26.12.

In tight junctions, regulates the reaction rate while the tight junction interacts with the reaction rate at step 26.13.

In tight junctions, limits the transport pathway while the tight junction interacts with the surrounding fluid at step 26.14.

In tight junctions, limits nearby organelles while the tight junction interacts with the surrounding cytoplasm at step 26.15.

In tight junctions, regulates the surrounding fluid while the tight junction interacts with nearby organelles at step 26.16.

In tight junctions, supports the transport pathway while the tight junction interacts with the surrounding fluid at step 26.17.

In tight junctions, stabilises nearby organelles while the tight junction interacts with the reaction rate at step 26.18.

In tight junctions, limits the transport pathway while the tight junction interacts with the surrounding cytoplasm at step 26.19.

In tight junctions, regulates the surrounding fluid while the tight junction interacts with nearby organelles at step 26.20.

In tight junctions, stabilises adjacent tissue while the tight junction interacts with the surrounding fluid at step 26.21.

In tight junctions, regulates nearby organelles while the tight junction interacts with the surrounding cytoplasm at step 26.22.

In tight junctions, regulates the surrounding fluid while the tight junction interacts with the surrounding fluid at step 26.23.

In tight junctions, regulates the reaction rate while the tight junction interacts with the surrounding cytoplasm at step 26.24.

In tight junctions, stabilises the reaction rate while the tight junction interacts with the surrounding cytoplasm at step 26.25.

In tight junctions, stabilises nearby organelles while the tight junction interacts with the surrounding cytoplasm at step 26.26.

In tight junctions, controls the surrounding fluid while the tight junction interacts with the transport pathway at step 26.27.

In tight junctions, stabilises nearby organelles while the tight junction interacts with the surrounding fluid at step 26.28.

In tight junctions, coordinates adjacent tissue while the tight junction interacts with the surrounding cytoplasm at step 26.29.

In tight junctions, controls the reaction rate while the tight junction interacts with the surrounding fluid at step 26.30.

In tight junctions, coordinates the surrounding fluid while the tight junction interacts with the reaction rate at step 26.31.

In tight junctions, stabilises the surrounding fluid while the tight junction interacts with the reaction rate at step 26.32.

In tight junctions, regulates the internal gradient while the tight junction interacts with the surrounding fluid at step 26.33.

27. Gap Junctions

Gap junction is defined as a channel that allows small molecules to pass between cells.

As a consequence of the fact that gap junction limits the surrounding fluid, the cell maintains its function.

The gap junction differs from the internal gradient in both structure and role.

During this stage the gap junction regulates the surrounding fluid.

In gap junctions, regulates the surrounding cytoplasm while the gap junction interacts with the surrounding cytoplasm at step 27.1.

In gap junctions, regulates the surrounding fluid while the gap junction interacts with the surrounding fluid at step 27.2.

In gap junctions, modulates the transport pathway while the gap junction interacts with the surrounding cytoplasm at step 27.3.

In gap junctions, limits the internal gradient while the gap junction interacts with the internal gradient at step 27.3.

In gap junctions, coordinates the transport pathway while the gap junction interacts with adjacent tissue at step 27.4.

In gap junctions, modulates the reaction rate while the gap junction interacts with nearby organelles at step 27.5.

In gap junctions, supports the surrounding cytoplasm while the gap junction interacts with the internal gradient at step 27.6.

In gap junctions, controls the transport pathway while the gap junction interacts with the surrounding fluid at step 27.7.

In gap junctions, limits nearby organelles while the gap junction interacts with the surrounding fluid at step 27.8.

In gap junctions, stabilises adjacent tissue while the gap junction interacts with the reaction rate at step 27.9.

In gap junctions, regulates the internal gradient while the gap junction interacts with the surrounding fluid at step 27.10.

In gap junctions, stabilises the surrounding fluid while the gap junction interacts with the reaction rate at step 27.11.

In gap junctions, limits nearby organelles while the gap junction interacts with nearby organelles at step 27.12.

In gap junctions, modulates the reaction rate while the gap junction interacts with nearby organelles at step 27.13.

In gap junctions, controls the reaction rate while the gap junction interacts with the reaction rate at step 27.14.

In gap junctions, supports the internal gradient while the gap junction interacts with the internal gradient at step 27.15.

In gap junctions, controls the surrounding cytoplasm while the gap junction interacts with the transport pathway at step 27.16.

In gap junctions, coordinates the surrounding fluid while the gap junction interacts with the reaction rate at step 27.17.

In gap junctions, modulates the transport pathway while the gap junction interacts with the internal gradient at step 27.18.

In gap junctions, modulates the transport pathway while the gap junction interacts with the surrounding fluid at step 27.19.

In gap junctions, regulates the reaction rate while the gap junction interacts with adjacent tissue at step 27.20.

In gap junctions, supports the reaction rate while the gap junction interacts with the internal gradient at step 27.21.

In gap junctions, supports nearby organelles while the gap junction interacts with adjacent tissue at step 27.22.

In gap junctions, limits nearby organelles while the gap junction interacts with the surrounding fluid at step 27.23.

In gap junctions, limits the transport pathway while the gap junction interacts with the reaction rate at step 27.24.

In gap junctions, stabilises the reaction rate while the gap junction interacts with nearby organelles at step 27.25.

In gap junctions, controls the surrounding fluid while the gap junction interacts with the surrounding cytoplasm at step 27.26.

In gap junctions, controls the internal gradient while the gap junction interacts with the internal gradient at step 27.27.

In gap junctions, limits adjacent tissue while the gap junction interacts with the transport pathway at step 27.28.

In gap junctions, modulates the reaction rate while the gap junction interacts with the reaction rate at step 27.29.

In gap junctions, regulates the internal gradient while the gap junction interacts with the reaction rate at step 27.30.

In gap junctions, stabilises the reaction rate while the gap junction interacts with the reaction rate at step 27.31.

In gap junctions, supports adjacent tissue while the gap junction interacts with the internal gradient at step 27.32.

In gap junctions, modulates the reaction rate while the gap junction interacts with the reaction rate at step 27.33.

28. Chromosomes

Chromosome is defined as a condensed structure of DNA that carries genes.

As a consequence of the fact that chromosome controls the transport pathway, the cell maintains its function.

The chromosome differs from the surrounding cytoplasm in both structure and role.

During this stage the chromosome supports the transport pathway.

In chromosomes, limits the reaction rate while the chromosome interacts with nearby organelles at step 28.0.

In chromosomes, stabilises the reaction rate while the chromosome interacts with the reaction rate at step 28.1.

In chromosomes, coordinates adjacent tissue while the chromosome interacts with the internal gradient at step 28.

In chromosomes, supports the surrounding cytoplasm while the chromosome interacts with the surrounding fluid at

In chromosomes, limits nearby organelles while the chromosome interacts with the surrounding fluid at step 28.4.

In chromosomes, stabilises the internal gradient while the chromosome interacts with the transport pathway at step

In chromosomes, modulates the surrounding cytoplasm while the chromosome interacts with the reaction rate at st

In chromosomes, limits nearby organelles while the chromosome interacts with the transport pathway at step 28.7.

In chromosomes, regulates the transport pathway while the chromosome interacts with the reaction rate at step 28

In chromosomes, controls the reaction rate while the chromosome interacts with the surrounding cytoplasm at step

In chromosomes, stabilises nearby organelles while the chromosome interacts with the transport pathway at step 2

In chromosomes, supports the internal gradient while the chromosome interacts with the reaction rate at step 28.11

In chromosomes, supports the internal gradient while the chromosome interacts with nearby organelles at step 28.

In chromosomes, supports the transport pathway while the chromosome interacts with adjacent tissue at step 28.1

In chromosomes, stabilises adjacent tissue while the chromosome interacts with adjacent tissue at step 28.14.

In chromosomes, regulates adjacent tissue while the chromosome interacts with the reaction rate at step 28.15.

In chromosomes, coordinates the internal gradient while the chromosome interacts with the internal gradient at ste

In chromosomes, supports the transport pathway while the chromosome interacts with the internal gradient at step

In chromosomes, limits the reaction rate while the chromosome interacts with the transport pathway at step 28.18.

In chromosomes, modulates adjacent tissue while the chromosome interacts with adjacent tissue at step 28.19.

In chromosomes, regulates nearby organelles while the chromosome interacts with the internal gradient at step 28

In chromosomes, modulates the surrounding cytoplasm while the chromosome interacts with the internal gradient a

In chromosomes, coordinates nearby organelles while the chromosome interacts with the reaction rate at step 28.2

In chromosomes, regulates adjacent tissue while the chromosome interacts with the internal gradient at step 28.23

In chromosomes, supports the surrounding cytoplasm while the chromosome interacts with the internal gradient at

In chromosomes, controls the transport pathway while the chromosome interacts with the transport pathway at step

In chromosomes, regulates the surrounding cytoplasm while the chromosome interacts with the surrounding cytopl

In chromosomes, stabilises the reaction rate while the chromosome interacts with the reaction rate at step 28.27.

In chromosomes, supports nearby organelles while the chromosome interacts with the transport pathway at step 2

In chromosomes, supports adjacent tissue while the chromosome interacts with the internal gradient at step 28.29.

In chromosomes, modulates the internal gradient while the chromosome interacts with nearby organelles at step 2

In chromosomes, regulates nearby organelles while the chromosome interacts with the reaction rate at step 28.31.

In chromosomes, supports the reaction rate while the chromosome interacts with the transport pathway at step 28.

In chromosomes, stabilises the internal gradient while the chromosome interacts with the reaction rate at step 28.3

29. Genes

Gene is defined as a segment of DNA that encodes a functional product.

Because gene controls adjacent tissue, the cell maintains its function.

The gene differs from adjacent tissue in both structure and role.

During this stage the gene limits the surrounding cytoplasm.

In genes, regulates the surrounding cytoplasm while the gene interacts with the surrounding cytoplasm at step 29.0.

In genes, supports the internal gradient while the gene interacts with the reaction rate at step 29.1.

In genes, coordinates nearby organelles while the gene interacts with nearby organelles at step 29.2.

In genes, modulates the surrounding cytoplasm while the gene interacts with the reaction rate at step 29.3.

In genes, supports the surrounding fluid while the gene interacts with nearby organelles at step 29.4.

In genes, regulates the surrounding fluid while the gene interacts with the surrounding cytoplasm at step 29.5.

In genes, controls the internal gradient while the gene interacts with the transport pathway at step 29.6.

In genes, stabilises the surrounding fluid while the gene interacts with the surrounding cytoplasm at step 29.7.

In genes, coordinates the transport pathway while the gene interacts with nearby organelles at step 29.8.

In genes, stabilises nearby organelles while the gene interacts with the reaction rate at step 29.9.

In genes, stabilises the internal gradient while the gene interacts with adjacent tissue at step 29.10.

In genes, coordinates adjacent tissue while the gene interacts with adjacent tissue at step 29.11.

In genes, regulates the internal gradient while the gene interacts with the internal gradient at step 29.12.

In genes, regulates the transport pathway while the gene interacts with the surrounding cytoplasm at step 29.13.

In genes, modulates the surrounding cytoplasm while the gene interacts with the internal gradient at step 29.14.

In genes, modulates the transport pathway while the gene interacts with the surrounding fluid at step 29.15.

In genes, coordinates the surrounding fluid while the gene interacts with the reaction rate at step 29.16.

In genes, limits the surrounding cytoplasm while the gene interacts with the surrounding cytoplasm at step 29.17.

In genes, stabilises the internal gradient while the gene interacts with the reaction rate at step 29.18.

In genes, regulates adjacent tissue while the gene interacts with the surrounding fluid at step 29.19.

In genes, coordinates the internal gradient while the gene interacts with the transport pathway at step 29.20.

In genes, supports nearby organelles while the gene interacts with the reaction rate at step 29.21.

In genes, coordinates the surrounding cytoplasm while the gene interacts with nearby organelles at step 29.22.

In genes, controls the internal gradient while the gene interacts with the internal gradient at step 29.23.

In genes, limits the surrounding cytoplasm while the gene interacts with the internal gradient at step 29.24.

In genes, limits nearby organelles while the gene interacts with adjacent tissue at step 29.25.

In genes, limits adjacent tissue while the gene interacts with the reaction rate at step 29.26.

In genes, stabilises the surrounding fluid while the gene interacts with the surrounding cytoplasm at step 29.27.

In genes, limits the surrounding fluid while the gene interacts with adjacent tissue at step 29.28.

In genes, modulates the surrounding cytoplasm while the gene interacts with adjacent tissue at step 29.29.

In genes, modulates adjacent tissue while the gene interacts with the surrounding cytoplasm at step 29.30.

In genes, supports the reaction rate while the gene interacts with the internal gradient at step 29.31.

In genes, coordinates adjacent tissue while the gene interacts with the reaction rate at step 29.32.

In genes, limits the surrounding cytoplasm while the gene interacts with nearby organelles at step 29.33.

30. Operons

Operon is defined as a cluster of genes transcribed together under a single promoter.
Because operon coordinates the surrounding cytoplasm, the cell maintains its function.
The operon differs from nearby organelles in both structure and role.

During this stage the operon controls the internal gradient.

In operons, modulates adjacent tissue while the operon interacts with nearby organelles at step 30.0.

In operons, regulates the surrounding fluid while the operon interacts with the internal gradient at step 30.1.

In operons, stabilises the internal gradient while the operon interacts with adjacent tissue at step 30.2.

In operons, coordinates the surrounding cytoplasm while the operon interacts with adjacent tissue at step 30.3.

In operons, coordinates nearby organelles while the operon interacts with the transport pathway at step 30.4.

In operons, stabilises nearby organelles while the operon interacts with the reaction rate at step 30.5.

In operons, stabilises the internal gradient while the operon interacts with nearby organelles at step 30.6.

In operons, limits adjacent tissue while the operon interacts with adjacent tissue at step 30.7.

In operons, regulates the internal gradient while the operon interacts with the transport pathway at step 30.8.

In operons, modulates the internal gradient while the operon interacts with the internal gradient at step 30.9.

In operons, modulates adjacent tissue while the operon interacts with the internal gradient at step 30.10.

In operons, limits the surrounding cytoplasm while the operon interacts with the internal gradient at step 30.11.

In operons, limits nearby organelles while the operon interacts with the surrounding cytoplasm at step 30.12.

In operons, stabilises the transport pathway while the operon interacts with the surrounding cytoplasm at step 30.13.

In operons, coordinates the reaction rate while the operon interacts with the surrounding fluid at step 30.14.

In operons, stabilises the transport pathway while the operon interacts with nearby organelles at step 30.15.

In operons, modulates the internal gradient while the operon interacts with the surrounding cytoplasm at step 30.16.

In operons, controls the reaction rate while the operon interacts with adjacent tissue at step 30.17.

In operons, controls nearby organelles while the operon interacts with the internal gradient at step 30.18.

In operons, stabilises adjacent tissue while the operon interacts with the surrounding cytoplasm at step 30.19.

In operons, limits the internal gradient while the operon interacts with nearby organelles at step 30.20.

In operons, stabilises the surrounding cytoplasm while the operon interacts with the reaction rate at step 30.21.

In operons, coordinates the internal gradient while the operon interacts with adjacent tissue at step 30.22.

In operons, coordinates the surrounding cytoplasm while the operon interacts with nearby organelles at step 30.23.

In operons, modulates the transport pathway while the operon interacts with the internal gradient at step 30.24.

In operons, supports adjacent tissue while the operon interacts with nearby organelles at step 30.25.

In operons, regulates the reaction rate while the operon interacts with the reaction rate at step 30.26.

In operons, supports the internal gradient while the operon interacts with adjacent tissue at step 30.27.

In operons, controls nearby organelles while the operon interacts with the surrounding cytoplasm at step 30.28.

In operons, limits adjacent tissue while the operon interacts with the internal gradient at step 30.29.

In operons, supports adjacent tissue while the operon interacts with adjacent tissue at step 30.30.

In operons, modulates adjacent tissue while the operon interacts with the transport pathway at step 30.31.

In operons, modulates adjacent tissue while the operon interacts with the surrounding fluid at step 30.32.

In operons, stabilises adjacent tissue while the operon interacts with the transport pathway at step 30.33.

References

Alberts B. Molecular Biology of the Cell.